

TECHNICAL MANUAL



LAS 5000XD

Cross Duct Tunable Laser Gas Analyzer

– DECEMBER 2022 –



WARNING

The information in this documentation is subject to change without notice.

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This document does not represent a commitment on the part of ENVEA.



WARNING!
INVISIBLE LASER RADIATION!

The LAS 5000XD uses lasers emitting light in the near-infrared region of the spectrum. When installed on a duct or stack, the LAS 5000XD is classified as a Class 3R or 3B laser product.



WARNING!
QUALIFIED PERSONNEL ONLY!

Electrical work must be carried out only by suitably trained and qualified personnel.



WARNING!
AUTHORISED PERSONNEL ONLY!

There are no user-serviceable parts inside the LAS 5000XD. Do not open the LAS 5000XD's enclosures unless authorized by the appropriate ENVEA representatives.

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Limits of use

To achieve optimum performance and safe operation of the LAS 5000XD, keep within the limits detailed in the technical specifications section of this manual (Section 1.2). Operation outside of these limits can result in failure to achieve the performance required and can cause damage to the equipment.

Additional information

Please contact ENVEA or your local ENVEA representative for information on product options, spares kits, special tools and accessories for the LAS 5000XD.

Product serial number

The serial number label is on the side of the Transmitter and Receiver. The serial number on the Transmitter and Receiver should match those on the corresponding test record for your LAS 5000XD.

1. GENERAL - CHARACTERISTICS

Read this manual before installing the LAS 5000XD. Follow all instructions to ensure that the LAS 5000XD works to specification and to avoid damaging the LAS 5000XD.

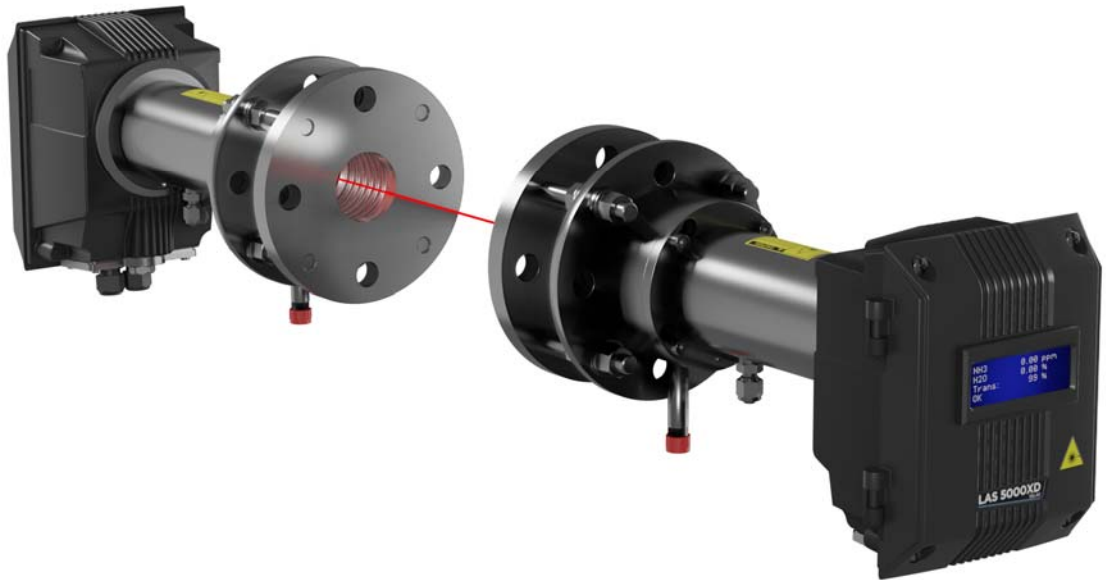


Figure 1-1 – Presentation of the LAS 5000XD

1.1. GENERAL CHARACTERISTICS

1.1.1. PRESENTATION

The LAS 5000XD is a cross-duct laser analyzer for monitoring concentrations of gases such as NH_3 and H_2O , HCl and H_2O , HF , CO and H_2O , CO_2 and H_2O or O_2 . Other gases are available on request.

The LAS 5000XD monitors gas concentration quantitatively using Tunable Diode Laser Absorption Spectroscopy.

The LAS 5000XD measures the target gas directly in a duct or stack. Therefore, there is no need for sample extraction or conditioning, and no need to transport gas to the LAS 5000XD.

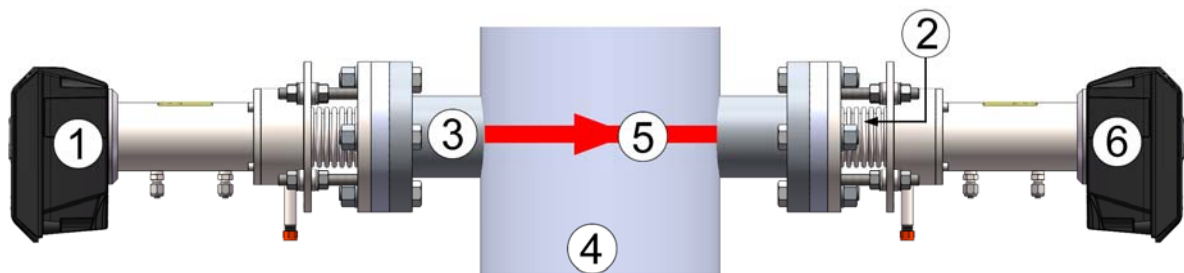
With its advanced optical and electronic technologies, the LAS 5000XD offers cross-duct monitoring with absolute accuracy. No calibration is necessary as the LAS 5000XD measures gas concentration directly from the recorded absorption spectrum using known molecular parameters. The LAS 5000XD offers stable and repeatable measurements over many years while requiring only limited maintenance.

The LAS 5000XD reports gas concentration using a 4-20 mA analog output or Modbus RTU. An LCD displays gas concentration and LAS 5000XD status. Diagnostic data is also available along with optional process temperature and pressure measurements.

The LAS 5000XD uses Tunable Diode Laser Absorption Spectroscopy to monitor gas concentrations in-situ.

1.1.2. DESCRIPTION

The overview of LAS 5000XD is shown below :



(1) Transmitter, (2) Alignment bellows, (3) Duct flange, (4) Process gas duct (cross section), (5) Laser light, (6) Receiver.

Figure 1–2 – LAS 5000XD overview

NOTE : Cross section showing laser beam passing through process gas is a representative image only. Duct dimensions can vary.

The basic working method of LAS 5000XD on a duct is as follows:

- The transmitter sends the laser light,
- The laser light passes through the process gas inside the duct,
- The receiver collects the laser light.

1.1.3. OPERATING MODES

The LAS 5000XD has two operating modes: standard mode and align mode. At any time, the user can switch from the standard mode to the align mode by clicking on the fast mode button in the Alignment page of the Web Server.

1.1.3.1. Standard mode

- Normal operation, measurement of gas concentration reported at 1 Hz.
- Concentration values in ppm , g/m³ ; mg/m³ ; µg/m³ ; ppb ; ppt ; g/Nm³ ; mg/Nm³ ; µg/Nm³ ; ppm ; ppbm ; pptm; or %m.
- Diagnostic data available for logging.

1.1.3.2. Align mode

- Diagnostic data available for logging.
- Received optical power measurement reported at 5-10 Hz, allowing for real-time adjustment of alignment bellows.
- No concentration measurement made.

1.1.4. ASSOCIATED EQUIPMENTS (OPTIONAL)

- Alignment tool and light source.
- Air purge conditioning. It is an air filter regulator which may or may not be equipped with a flow control.
- In-line span check cell.
- Audit cell.
- ProController, NetController, used for data visualization and logging.
- Junction box.
- Thermal Shield.
- Weather protection cover.

Please contact ENVEA or your local ENVEA representative for information.

1.1.4.1. Junction box

The junction box is an additional optional box allowing to give additional analog inputs/ outputs through an ESTEL box.

The junction box includes the following elements (refer to Figure 1–3):

- Power / Modbus RTU cable (1),
- USB ESTEL cable (2),
- AC/DC Power supply (3). In : 100 – 240 VAC ⇔ Out : 24 VDC,
- Circuit breaker C4A (4),
- Mains power supply (100-240 VAC) input cable : phase terminal on grey din rail (5), neutral terminal on blue din rail (6), ground terminal on yellow green din rail (7),
- 24 VDC power supply : ground terminal on yellow green din rail (8),
- Feed-through junction block (9),
- ESTEL module (10),
- Connector identification label of the feed-through junction block (11),

- Terminal block label of ESTEL module for J3 and J2 connectors (12).
- 100-230 VAC mains connection cable to 24 VDC power supply (13).
- 24 VDC cable (14).
- 100-230 VDC mains cable (15).
- Available cable glands (16).
- GND of analyzer and ESTEL module (17).
- + 24VDC of analyzer and ESTEL module (18).

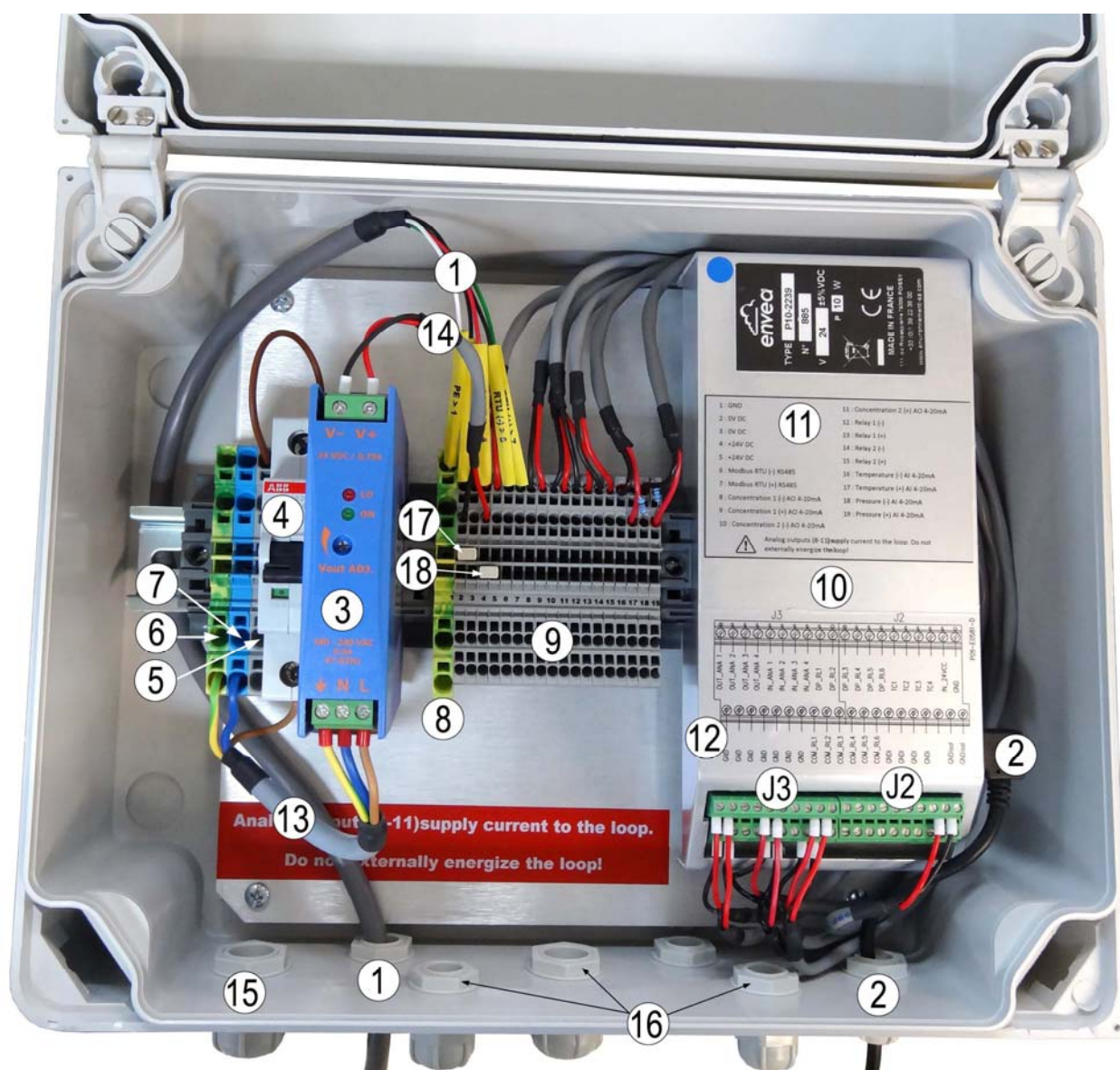


Figure 1-3 – Internal view of the Junction box



Analog outputs (8-11) supply current to the loop. DO NOT externally energize the loop.

Table 1-1 – Identification of AI/AO on J3 and J2 terminal block of ESTEL module

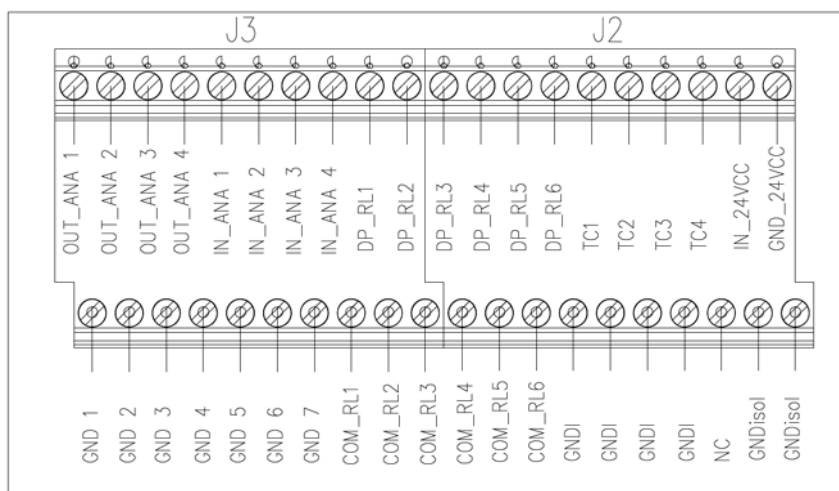


Table 1-2 – Connector identification of the feed-through junction block

Connector	Connection
1	GND
2	0V DC
3	0V DC
4	+24V DC
5	+24V DC
6	Modbus RTU (-) RS485
7	Modbus RTU (+) RS485
8	Concentration 1 (-) analog output 4-20 mA
9	Concentration 1 (+) analog output 4-20 mA
10	Concentration 2 (-) analog output 4-20 mA
11	Concentration 2 (+) analog output 4-20 mA
12	Relay 1 (-)
13	Relay 1 (+)
14	Relay 2 (-)
15	Relay 2 (+)
16	Temperature (+) analog input 4-20 mA
17	Temperature (+) analog input 4-20 mA
18	Pressure (-) analog input 4-20 mA
19	Pressure (+) analog input 4-20 mA



Analog outputs (8-11) supply current to the loop. DO NOT externally energize the loop.

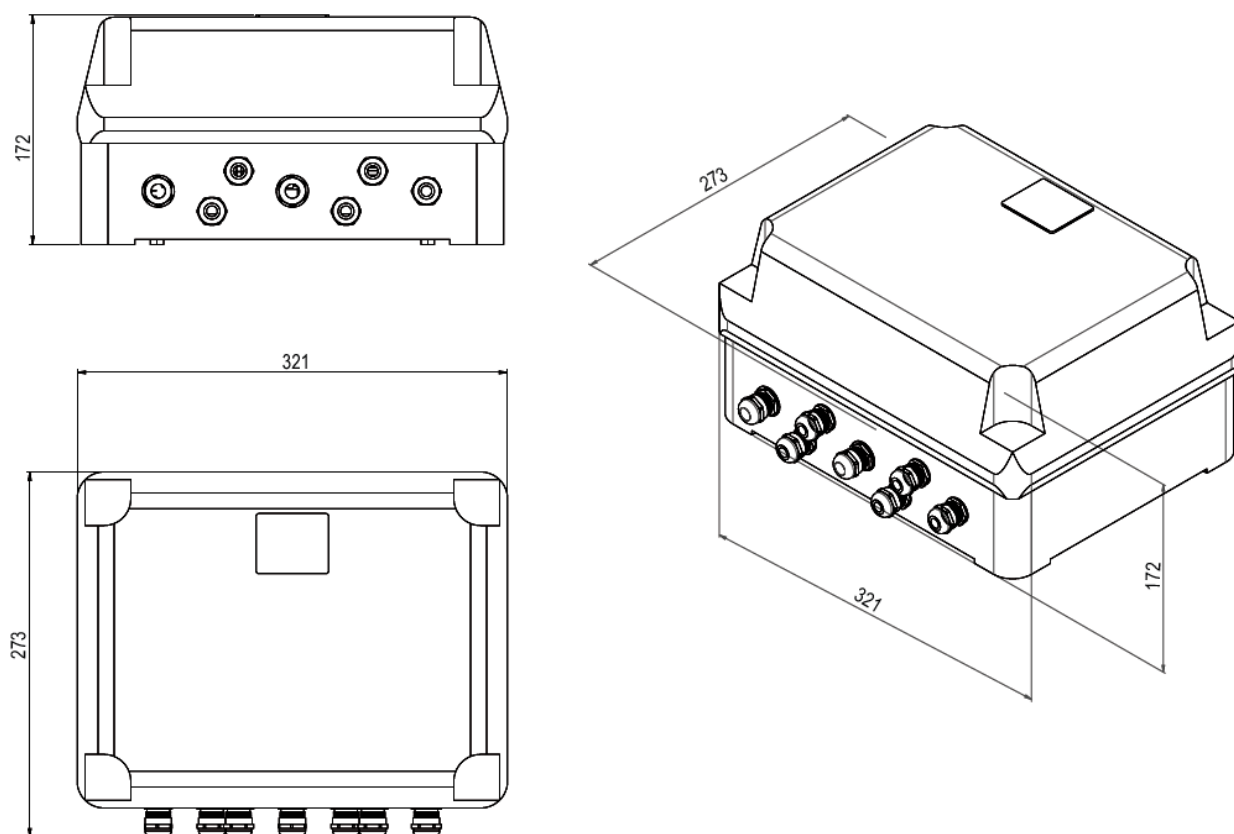
Junction box dimensions :

Length : 321 mm

Width : 273 mm

Height : 172 mm

Weight : 4.0 kg

**Figure 1-4 –Junction box dimensions**

1.1.4.2. Audit Cell

The Audit-Cell is presented below :



(1) Temperature sensor connection, (2) span gas inlet, (3) span gas outlet

Figure 1-5 – Audit-Cell presentation

The span gas inlet or outlet can be either on the transmitter side or the receiver side, the direction is not important.

If a pressure sensor is needed, it can be connected in series to (2) or (3).

After mounting the transmitter and the receiver on the audit-cell, the user should perform the alignment of the assembly.

Dimensions and weight of the Audit-Cell:

Length : 1324 mm

Height : 267 mm

Weight : 23 kg

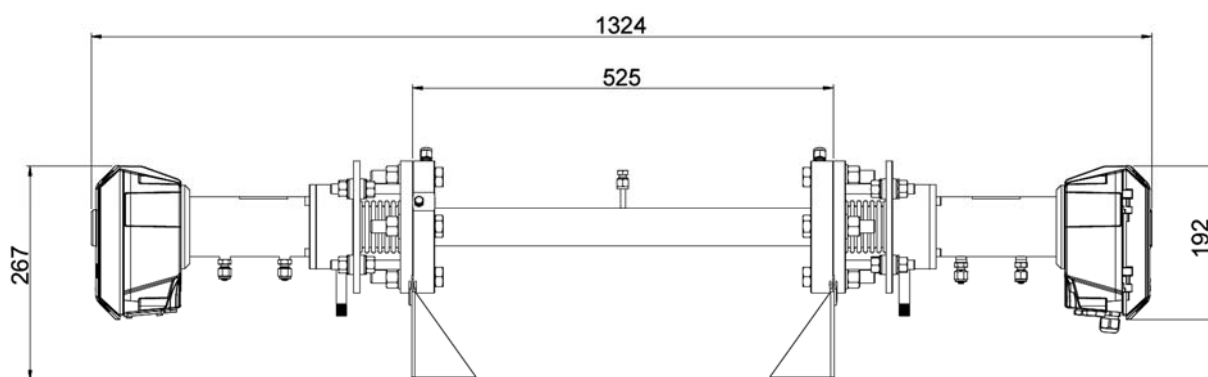
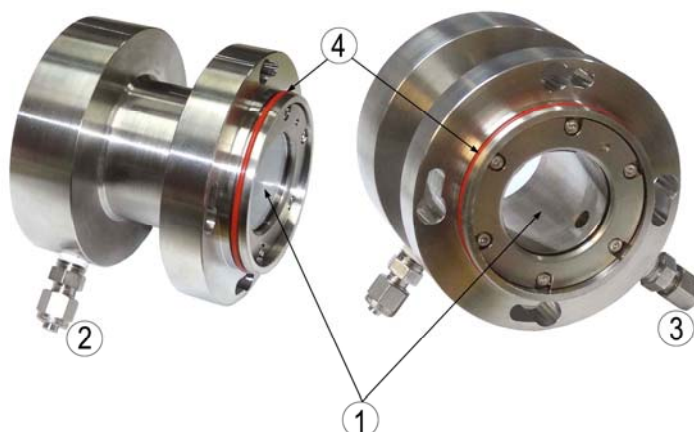


Figure 1-6 – Audit cell dimensions

1.1.4.3. In-line span check cell

The in-line span cell should be inserted between the Receiver and the alignment bellow. It is presented below :



(1) Window, (2) gas in, (3) gas out, (4) gasket

Figure 1-7 – Unfitted in-line span cell

Dimensions and weight of the In-line span cell:

Length	: 87 mm (when unfitted)
Length of the assembly	: 484 mm (receiver + in-line cell + alignment bellow)
Optical pathlength	: 104 mm
Weight, unfitted	: 3.1 kg

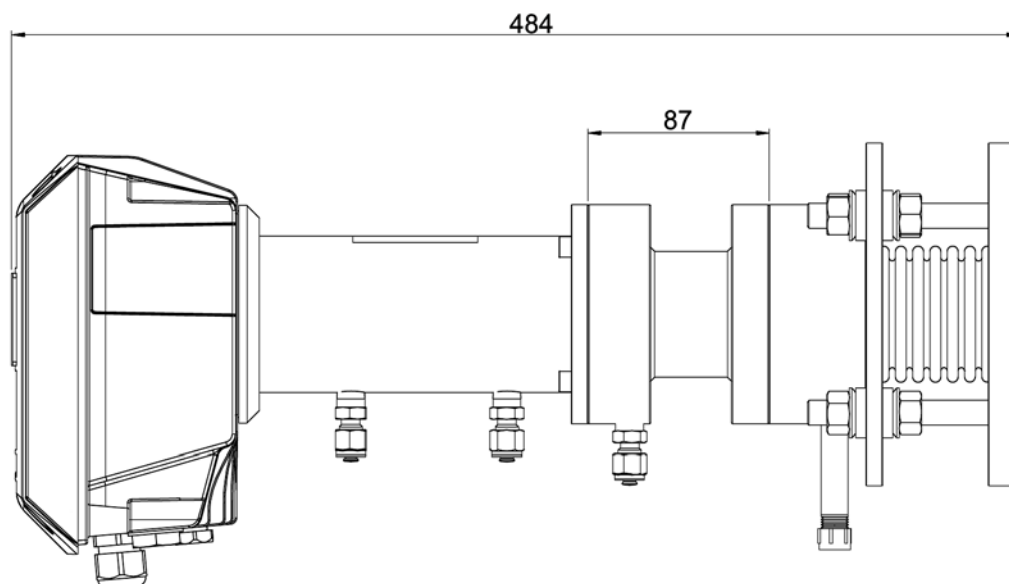


Figure 1-8 – In-line span cell dimensions

1.2. CHARACTERISTICS

1.2.1. TECHNICAL CHARACTERISTICS

The technical characteristics are defined in the following conditions :

- Gas temperature : 25°C
- Gas pressure : 1013 mbar
- Path length : 100 cm
- Ambient temperature : 25°C

1.2.1.1. Measurement ranges

H₂O and NH₃

- NH₃ : 0 – 10 ppm, 0 – 5000 ppm
- H₂O : 0 – 5 %, 0 – 40 %

HCl and H₂O

- HCl : coming soon
- H₂O : coming soon

HF : 0 – 3 ppm, 0 – 500 ppm

CO (ppm) : 0 – 50 ppm, 0 – 5000 ppm

CO (%) : 0 – 1 %, 0 – 100 %

O₂ : coming soon

CO₂ (%) : coming soon

Other gases : Available upon request

1.2.1.2. Sample gas maximum temperature

NH₃ + H₂O / HCl + H₂O / HF : +400°C (depends on the concentration range)

CO + H₂O / O₂ / CO + CO₂ : +1200°C (depends on the concentration range)

1.2.1.3. General specifications

Accuracy : $\leq \pm 1$ % of full scale

Response time (0–90 %) : 1 s

Zero drift : $\leq \pm 1$ % of full scale

Span drift : $\leq \pm 1$ % of full scale

Linearity : $\leq \pm 1$ % of full scale

Flue gas pressure : 2 bars max (absolute)

Display : 4 x 20 alphanumeric LED backlit LCD

Communication : Modbus RTU (RS485) / Ethernet (RJ45) - Web server

Input signals (Optional) : Temperature and pressure inputs (4-20 mA)

Output signals (Optional) : 2 analog outputs (4-20 mA), 2 relays

Power supply type : + 24V DC, ripple and noise 50 mV

Power consumption	: 15 W when starting-up the LAS 5000XD : < 15 W in normal operation
Recommended T°C	: - 20°C to + 55°C
IP index Tx & Rx enclosures	: IP65
Flange specification requirement on stack	: DN50 PN16, 2" – 150 lbs, Class 150.
Flange material	: SS 316 L
Air consumption (main purge - necessary)	: 5 - 50 L/ min (to be adjusted according to site conditions) (dry and oil free, ISO 8573.1 Class 2-3)
Air consumption (secondary purge – recommended)	: 2-3 L/min (dry and oil free, ISO 8573.1 Class 2-3)
Stack diameter compatibility	: From 0.5 to 20 m

1.2.1.4. Specifications of the supplied connecting cables

Power supply cable:	Provides + 24 V DC, GND and Protective earth, all must be connected. Standard lengths: 3 m
Ethernet cable:	A waterproof in-line connector is provided
Transmitter-Receiver cable:	Supplied attached to the Transmitter. It is not attached to the Receiver. Supplied in two standard lengths: 10 m and 30 m.
Input-output cable: (Supplied in the standard length of 3 m).	Contains the following signals: – External gas pressure sensor input (4-20mA analog connection). – External gas temperature sensor input (4-20mA analog connection). – Gas concentration 1 output (4-20mA analog connection). – Gas concentration 2 output (4-20mA analog connection). – Modbus RTU. – Relay outputs (normally open contacts).

NOTE : Cables must be connected to connection devices with protection mode listed in EN/IEC 60079-0.

1.2.2. STORAGE CHARACTERISTICS

- Temperature: -10°C to +60°C.

1.2.3. INSTALLATION CHARACTERISTICS**1.2.3.1. Dimensions and weight**

Refer to Figure 1–10.

The LAS 5000XD is composed of two elements, the transmitter and the receiver. Their dimensions are as follows:

Transmitter:

- Length: 401 mm (approximately 15.8 inches) excluding connectors and fittings on the rear panel.
- Width: 213 mm (approximately 8.4 inches)
- Height: 159 mm (approximately 6.25 inches)
- Weight: 9.5 kg with alignment bellow

Receiver:

- Length: 397 mm (approximately 15.6 inches) excluding connectors and fittings on the rear panel.
- Width: 215 mm (approximately 8,5 inches)
- Height: 159 mm (approximately 6.25 inches)
- Weight: 8.4 kg with alignment bellow

1.2.3.2. Handling and storage

The LAS 5000XD needs to be handle with care to avoid damage. If either the Transmitter or Receiver are removed from the alignment bellows, the exposed window aperture should be protected and kept clean. The exposed window location is indicated in Figure 1–9.

The LAS 5000XD should be stored in its original shipping material.

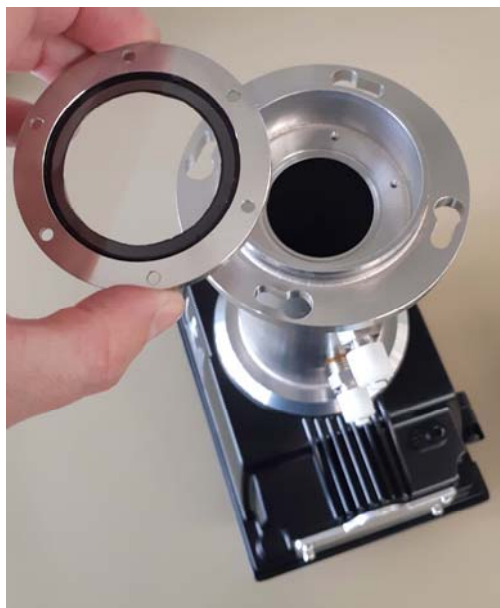
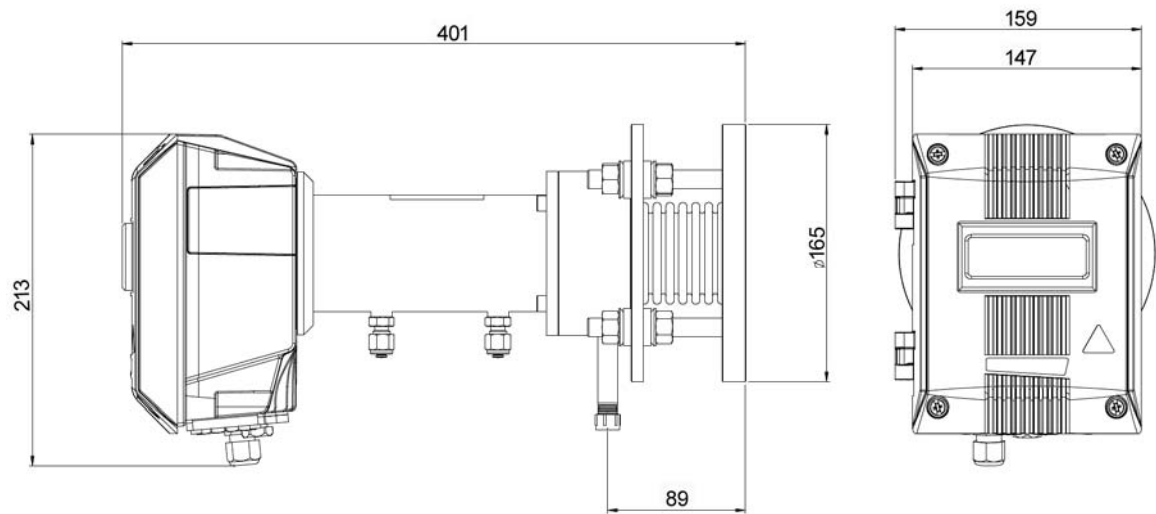
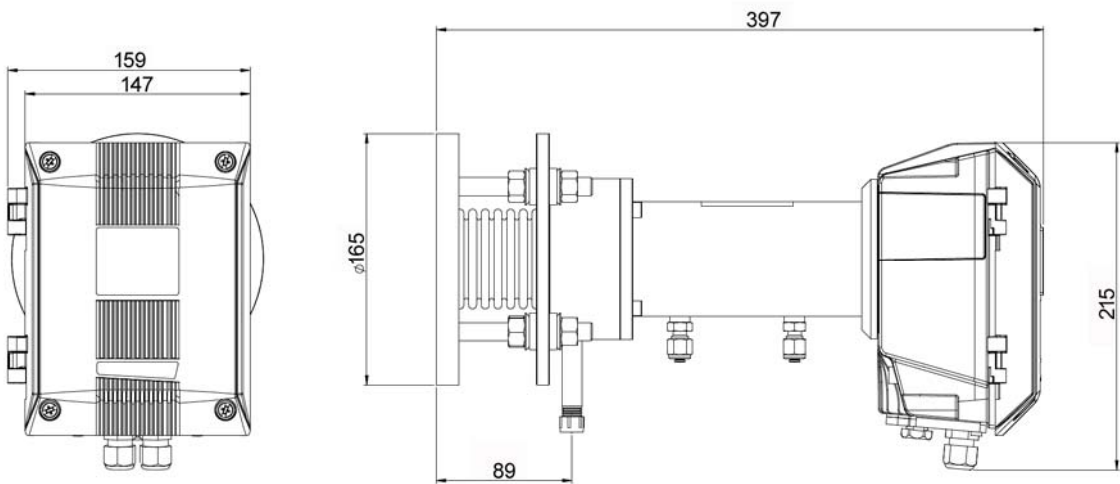


Figure 1–9 – Location of the exposed window



Transmitter dimensions



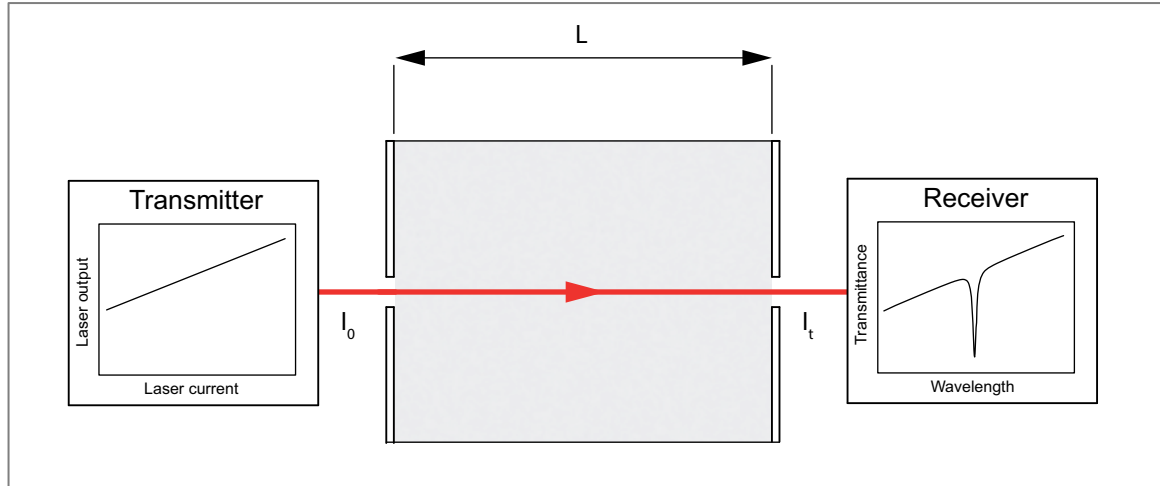
Receiver dimensions

Figure 1–10 – LAS 5000XD dimensions

2. PRINCIPLE OF OPERATION

The LAS 5000XD uses Tunable Diode Laser Absorption Spectroscopy for monitoring gases. The LAS 5000XD is a calibration-free gas monitoring system. The targeted molecule absorption line is rapidly scanned at rates of up to kHz frequencies.

The light emitted from a single-mode laser source is directed through the gas to be measured. As the laser frequency is tuned, the intensity of light received by the LAS 5000XD Receiver varies depending on the concentration, temperature and pressure of the targeted gas.



The laser light emitted is tuned by varying the current supplied to the laser. A wavelength range of a few nanometers is monitored, a range which includes absorption features of the monitored gas.

The absorption of light follows the Beer-Lambert law, shown in equation 1:

$$T(\lambda) = \left(\frac{I_t(\lambda)}{I_0(\lambda)} \right) = e^{[\alpha(\lambda, P, T) \times L]}$$

Equation 1 – Beer-Lambert Law

The Beer-Lambert law describes the fundamental measurement, where:

- $T(\lambda)$ is the transmittance, depending on the wavelength, λ
- $I_t(\lambda)$ is the received optical intensity, depending on the wavelength, λ
- $I_0(\lambda)$ is the transmitted optical intensity, depending on the wavelength, λ
- L is the pathlength through the absorbing medium (cm)
- $\alpha(\lambda, P, T)$ is the absorption coefficient, depending on the wavelength, λ , the pressure, P and the temperature, T . The absorption coefficient can be expressed as shown in equation 2:

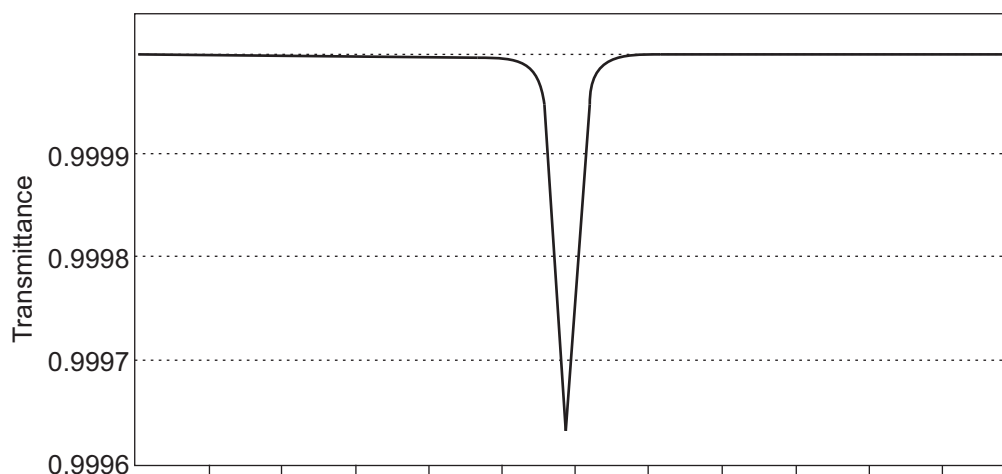
$$\alpha(\lambda, P, T) = S(\lambda, T) \times C \times f(\lambda, P, T)$$

Equation 2 – Absorption coefficient

Where:

- $S(\lambda, T)$ is the absorption line intensity (cm⁻¹/molecule.cm⁻²), depends on the wavelength, λ and temperature, T
- C is the concentration of the absorbing molecule (molecule/cm³)
- $f(\lambda, P, T)$ is the normalized line profile (cm), depends on the wavelength λ , pressure P and temperature T

Unlike other optical gas measurement techniques, the LAS 5000XD uses a full spectrum technique by tuning the laser wavelength over the absorption feature. This allows every measurement to use a normalised spectrum, ensuring that attenuation due to fouling on windows or by particulates do not affect the reported concentration. The following chart shows a typical normalised spectrum.



In the LAS 5000XD signal processing method, the linestrength is continuously adjusted for a measured gas temperature. Changes in pressure are also continuously normalised.

The absorption spectrum is processed by fitting the mathematically described peak lineshape to the recorded spectrum. This is a powerful pattern matching algorithm that finds the best fit of the absorption feature to the recorded spectrum. By applying the linestrength, temperature and pressure parameters to the results of this peak fitting procedure, an accurate gas concentration can be determined.

Absorption spectra contain a large amount of information about the gas being measured and, importantly, about the quality of the measurement, enabling the health of the LAS 5000XD to be continuously monitored. The standard deviation of the residual of the recorded spectrum and the fitted lineshape is a key indicator of the measurement quality. This measurement quality is obtained for every concentration measurement and is a reliable indicator of measurement confidence to the user.

3. OPERATING INSTRUCTIONS

3.1. INITIAL STARTUP

The LAS 5000XD is checked and calibrated in the factory before delivery. However, some parameters must be checked on-site when starting-up the analyzer to ensure correct operation.

3.1.1. CHOOSING THE INSTALLATION LOCATION



When choosing an installation location for the LAS 5000XD, it is necessary to make sure:

- There is enough room for the Transmitter and Receiver to be removed from the duct flange and for the engineer to have full access to the LAS 5000XD, its display and its cables.
- The supporting structure is appropriate for the weight of the LAS 5000XD and its cables.

Summary	Details
Laser beam must pass through process gas.	– Mount the LAS 5000XD in a part of the duct where the laser beam passes through the process gas and a representative measurement can be achieved.
Clear line-of-sight.	– Make sure there is a clear line-of-sight between the Transmitter and the Receiver through the duct.
Dust load.	– Choose a location where there is a low dust load or make provision for reducing the dust load along the laser path. For high dust loads, please consult ENVEA or your local ENVEA representative.
Pathlength 0.5 m to 20 m.	– The typical pathlength between the Transmitter and the Receiver is 0.5 m to 20 m. If the pathlength is longer than 5 m, please consult ENVEA or your local ENVEA representative.
Avoid excessive vibration.	

3.1.2. THE DUCT FLANGE

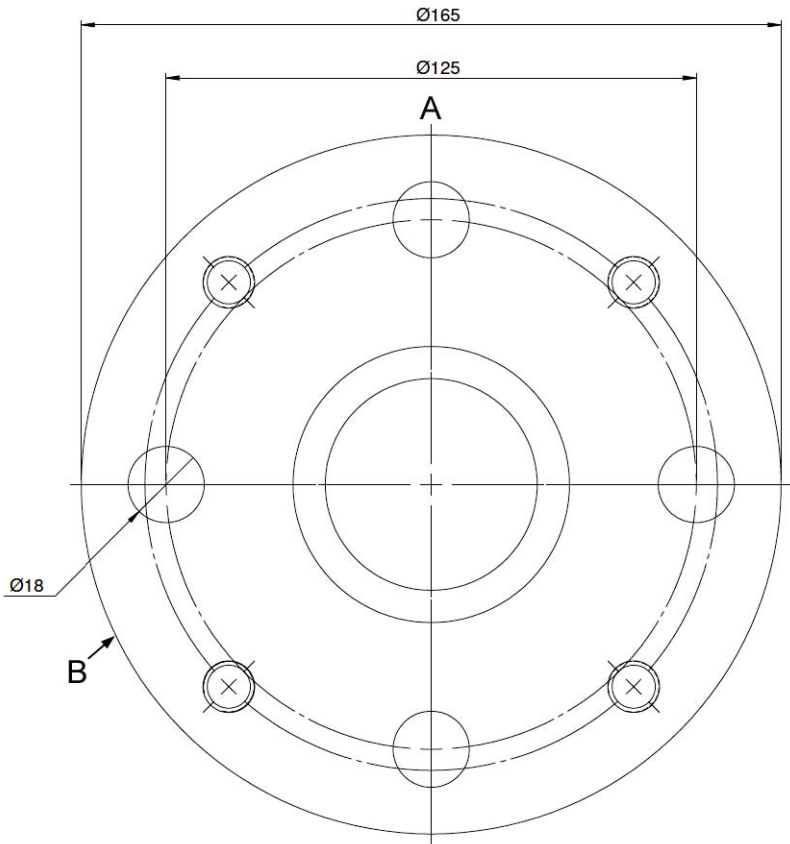


It is recommended that duct flange work is carried out when the process is off.

3.1.2.1. Duct flange requirements

The dimensions of the duct flange are the following (see Figure 3–1):

- External diameter of the duct flange: 165 mm
- Mounting hole diameter: 18 mm
- Positioning diameter of drill holes : 125 mm



Legends:

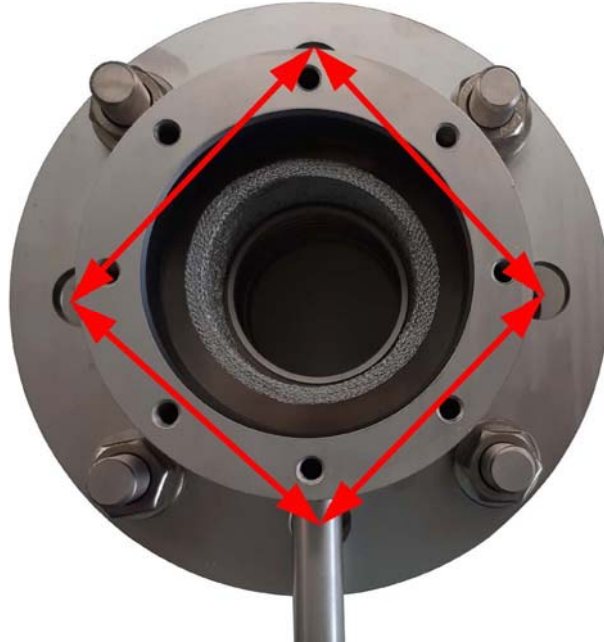
Key	Note
A	Flange should be orientated so that the mounting holes are as shown above with the top at A. For an easiest understanding, see Figure 3–2.
B	Duct flange adapter. Available from your supplier. Flange specification requirement on stack: DN50 PN16, 2” – 150 lbs, Class 150.

Figure 3–1 – Duct flange dimensions

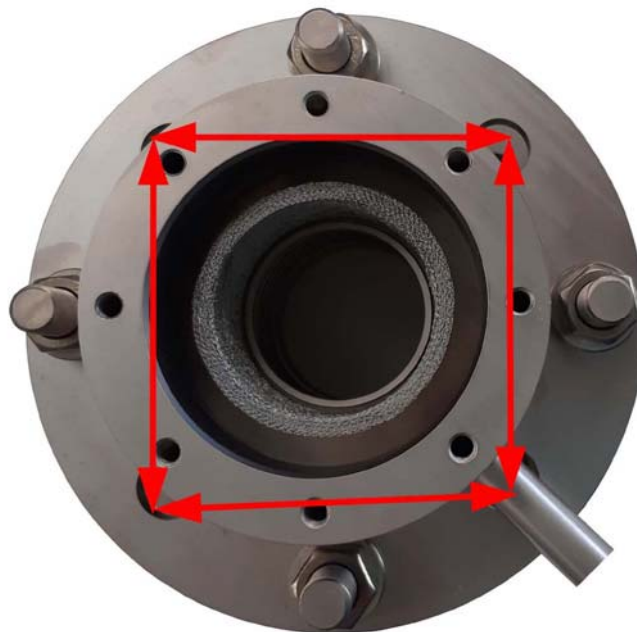


Pay attention to mount the flange with the holes oriented as rhomb.

If it is not the case, the analyzer will not be upright, and the optional weather protection cover cannot be mounted.



CORRECT MOUNTING : mounting holes oriented as rhomb.



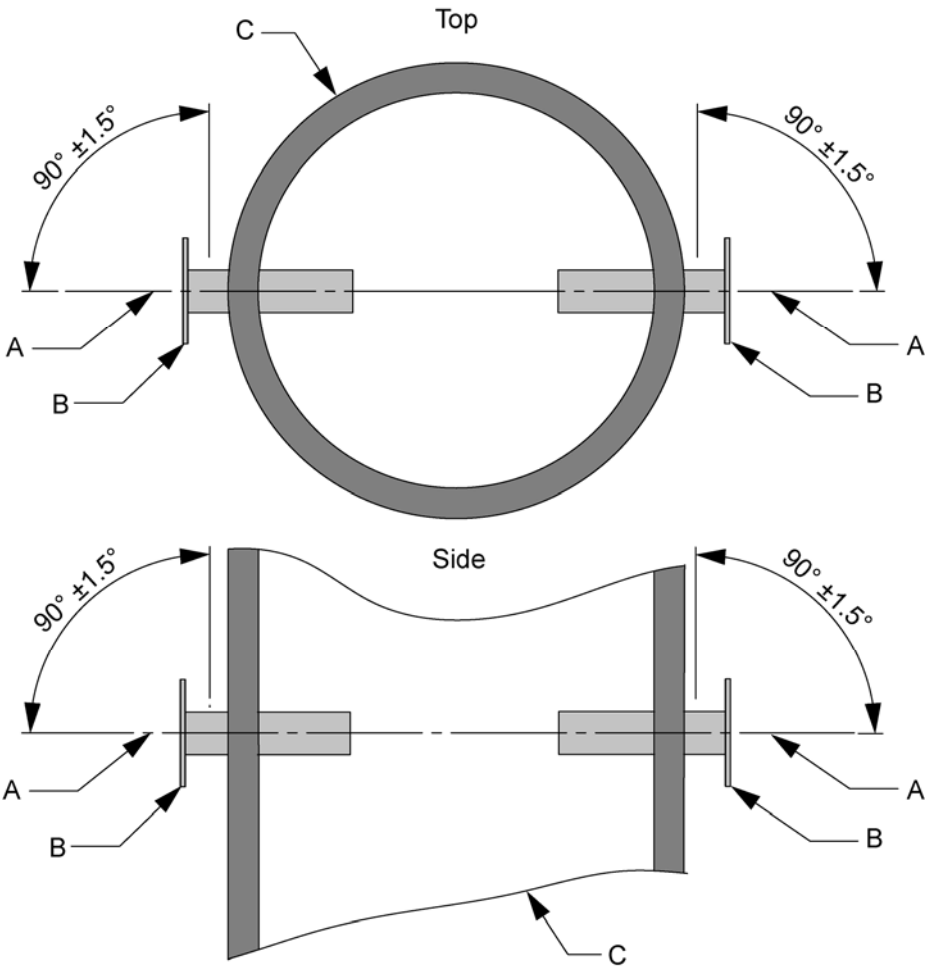
WRONG MOUNTING : mounting holes oriented as square.

Figure 3-2 – Mounting hole orientation of duct flange

3.1.2.2. Duct flange alignment – angle

The initial installation of the duct flanges is very important. The flanges should be aligned as precisely as possible. Consequently, the following installation recommendations should be respected:

- 
- The duct flanges should be in-line and parallel.
 - During duct flange installation, the user has to achieve the best possible position and aim for an angle better than $\pm 1.5^\circ$ as shown in the above figures.
 - He has to aim for a direct line-of-sight through the centre of the duct flanges.



Legends:

Key	Note
A	Centre of duct flange.
B	Duct flange.
C	Cross section of duct. Representative images: duct dimensions can vary.

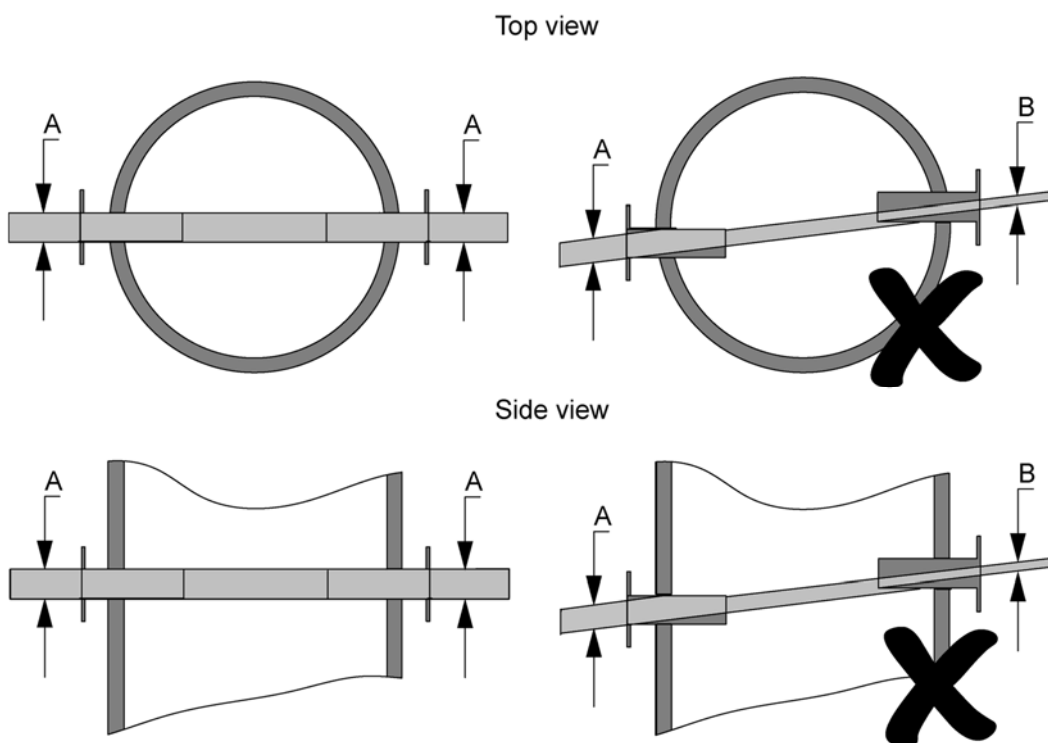
Figure 3-3 – Angle of duct flange alignment

3.1.2.3. Duct flange alignment – offset

It is recommended to avoid offsetting the duct flanges and to maximize the clear view from one aperture to the other.



If the duct flanges are offset, the maximum amount of light received by the Receiver will be limited, leading to poor LAS 5000XD performance.



Legend:

Key	Explanation	Result
A	Maximum path width for light because the duct flanges are not offset.	Good.
B	Reduced path width for light, caused by offsetting the flanges.	Unsuitable.

Figure 3–4 – Images of offset of duct flange alignment (*)

(*) Representative images: duct dimensions can vary.

3.1.3. INSTALLATION AND COMMISSIONING THE LAS 5000XD

3.1.3.1. Tools required

The tools required for the LAS 5000XD installation and commissioning are the following:

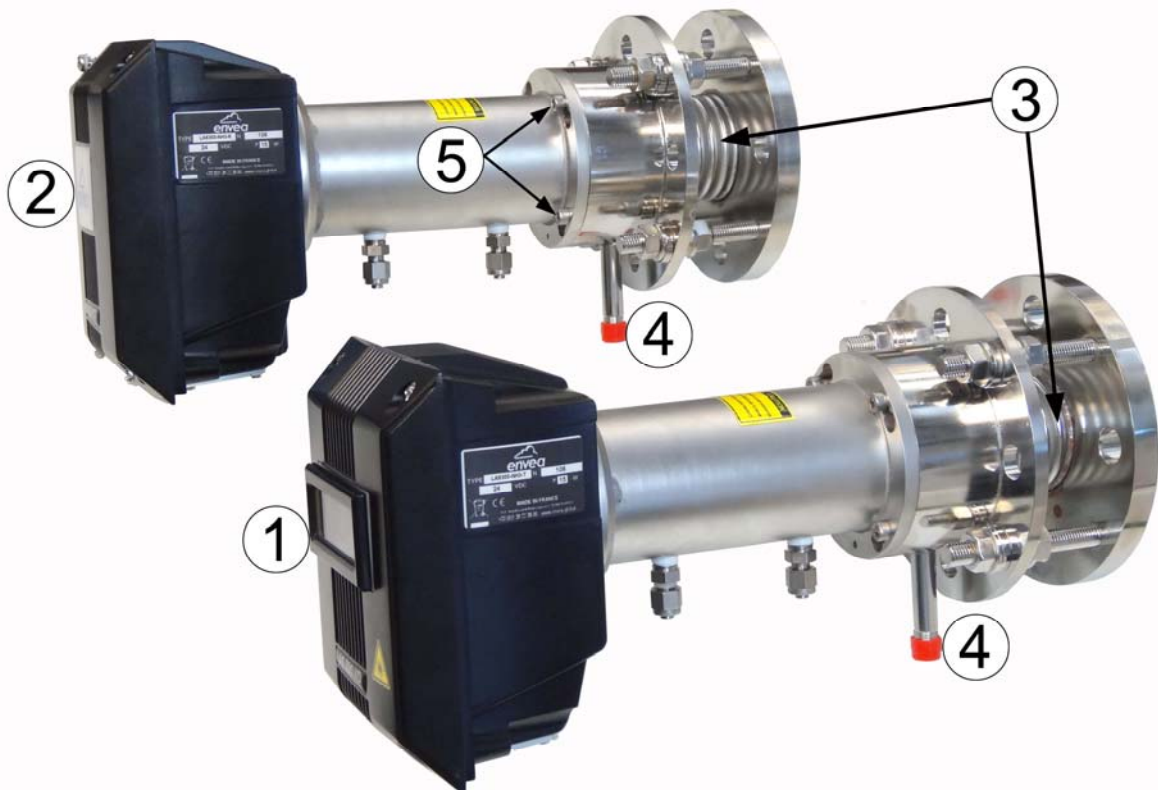
- 3 mm Allen key for the bolts that attach the cable gland plate, and 5 mm for the bolts on the alignment bellows.
- 19 mm spanner for the alignment bellows nuts, and 24 mm for mounting the alignment bellows to the flange.

First steps to be carried out, they are the following.

3.1.3.2. Preliminary mounting operations

Starting up the analyzer involves the following preliminary operations: refer to Figure 3–5.

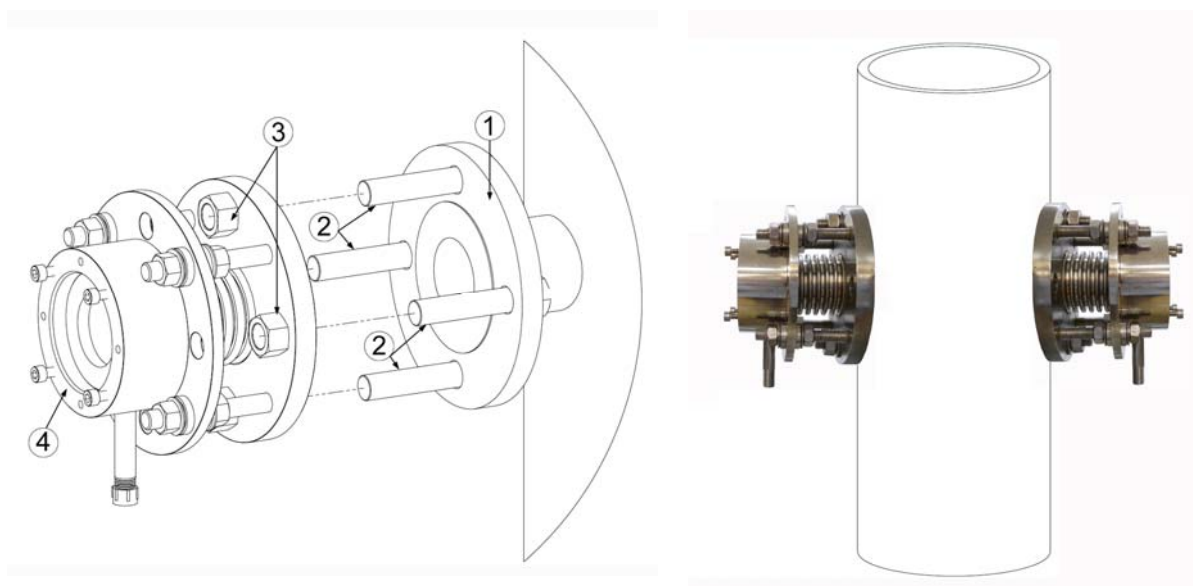
- Unpack the unit then and visually inspect the transmitter and the receiver to make sure no damage occurred during transport. The Transmitter (1) and Receiver (2) are supplied attached to alignment bellows (3).
- Separate the alignment bellow assembly from both the transmitter and the receiver:
 - Use a 5 mm Allen key to loosen the four bolts (5),
 - Pull the transmitter and receiver out of the alignment bellows assembly.
 - Carefully put the transmitter or receiver aside, making sure the exposed window is kept free of contamination.
- Connect the purge inlets (4) with air of 35-40l/min flow. The purge inlet has a G $\frac{1}{4}$ thread connection.



(1) Transmitter, (2) Receiver, (3) alignment bellows, (4) purge air connection, (5) bolts.

Figure 3–5 – Air purge connection of transmitter and dismounting from alignment bellows

- Once the purge is connected, mount the alignment bellows of transmitter and receiver onto each side of the duct flanges (1) by inserting each alignment bellow on the four screws (2) of each duct flange and loosening the M16 nuts (3).



(1) Duct flange, (2) screws, (3) M16 nuts, (4) alignment flange.

Figure 3-6 – Mounting the alignment bellows on the duct flange

General purging advices:

Constant gas purge:	A constant gas purge is necessary at all times to keep the LAS 5000XD process windows free of process dust and to maintain the maximum light throughput. The Transmitter and Receiver each require a controlled and independent window purge.
Type of purge gas:	Purging commonly uses nitrogen or instrument air. When measuring oxygen, a nitrogen purge is required. Contact ENVEA or your local ENVEA representative for oxygen measurement advice.
Clean purge gas:	Use a clean supply of purge gas, free from particulates, oil and water.
Appropriate purge gas:	The purge gas flow rate must be appropriate for your system. It depends on: – Dust load. – Velocity of the gas in the duct. – Dimensions of the flanges.
Flow rate:	For illustration, a typical flow rate of 5-50 litres per minute is required. The gas purge supply pressure is defined by the flow rate needed.

Contact ENVEA or your local ENVEA representative for purge information specific to your system. ENVEA can supply a suitable purge system.

Window purge setup:

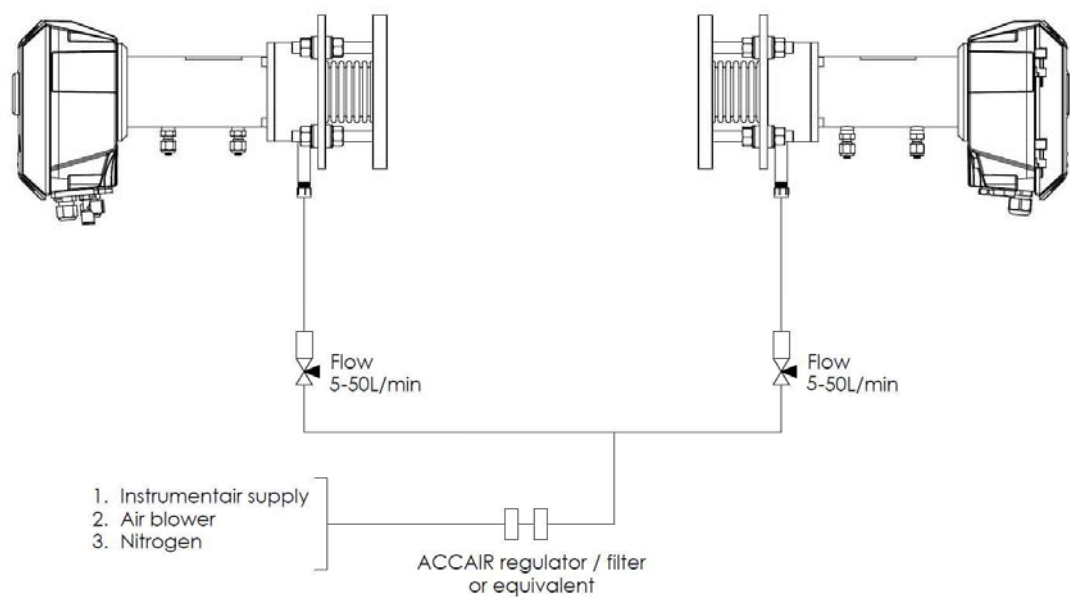
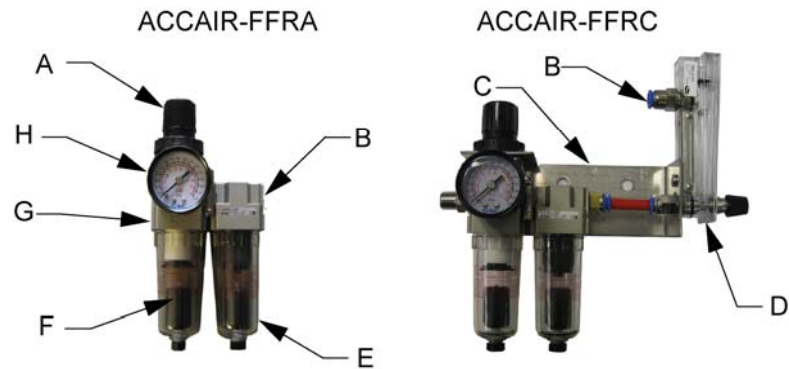


Figure 3–7 – Window purge setup

Available purge products:



A	Regulator to control delivery pressure
B	Air out (1/4" BSP)
C	Wall mounting bracket
D	Flow meter with control
E	Secondary filter for particulate >0.01 µm and oil
F	Pre-filter; nominal filter rating for particulate >5 µm
G	Air in (1/4" BSP)
H	Outlet air pressure gauge

ACCAIR-FFRA	(Left, in above image) ACCAIR-FFRA air filter regulator. Provides a constant flow of filtered instrument air to the LAS 5000XD. Requires additional flow control.
ACCAIR-FFRC	(Right, in above image) ACCAIR-FFRC air filter regulator with flow control. Provides a constant flow of filtered instrument air and also takes into account changes in duct pressure, adjusting the purge flow accordingly.
ACCAIR-B-Y blower	(Not pictured) Use if instrument air is not available. Supplies purge air, up to 50 litres/min, under normal duct conditions, typically -10 to +10 mbar.

The above purging products are available from ENVEA.

3.1.3.3. Preliminary alignment using alignment tools

Mounting the alignment tools on the alignment flanges:

The preliminary alignment has to be done with the alignment tools delivered with the transmitter and the receiver. These tools consist of two different discs equipped with a torch light (light source) (1) for one and an aiming device (2) for the other (see Figure 3–8).

The disc with the torch light should be mounted on the alignment flange on the transmitter side, and the disc with the aiming device on the alignment flange on the receiver side, following the procedure detailed below (refer to Figure 3–8 and Figure 3–9) :

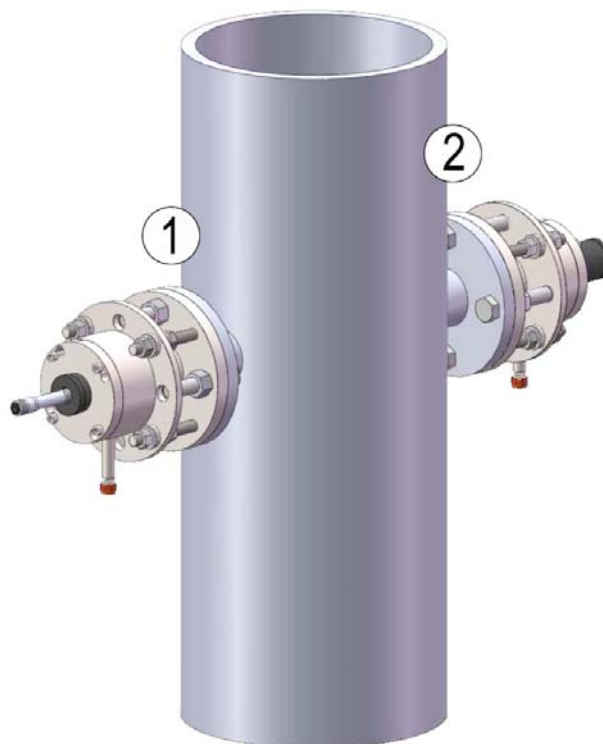
- Check that the light source is operational.
- Attach the light source on the Transmitter alignment flange (3) above the alignment bellows (4), and use a 5 mm Allen key to tighten the four bolts (5).
- Switch the light source on.
- Attach the aiming device on the Receiver alignment flange (6) above the alignment bellows (7), and use the 5 mm Allen key to tighten the four bolts (8).

NOTE : The source light is white and is of no danger.



(1) Torch light, (2) aiming device, (3) transmitter alignment flange, (4) alignment bellows, (5) bolts, (6) receiver alignment flange, (7) alignment bellows, (8) bolts.

Figure 3-8 – Mounting the alignment tools on the alignment flanges



(1) transmitter side, (2) receiver side.

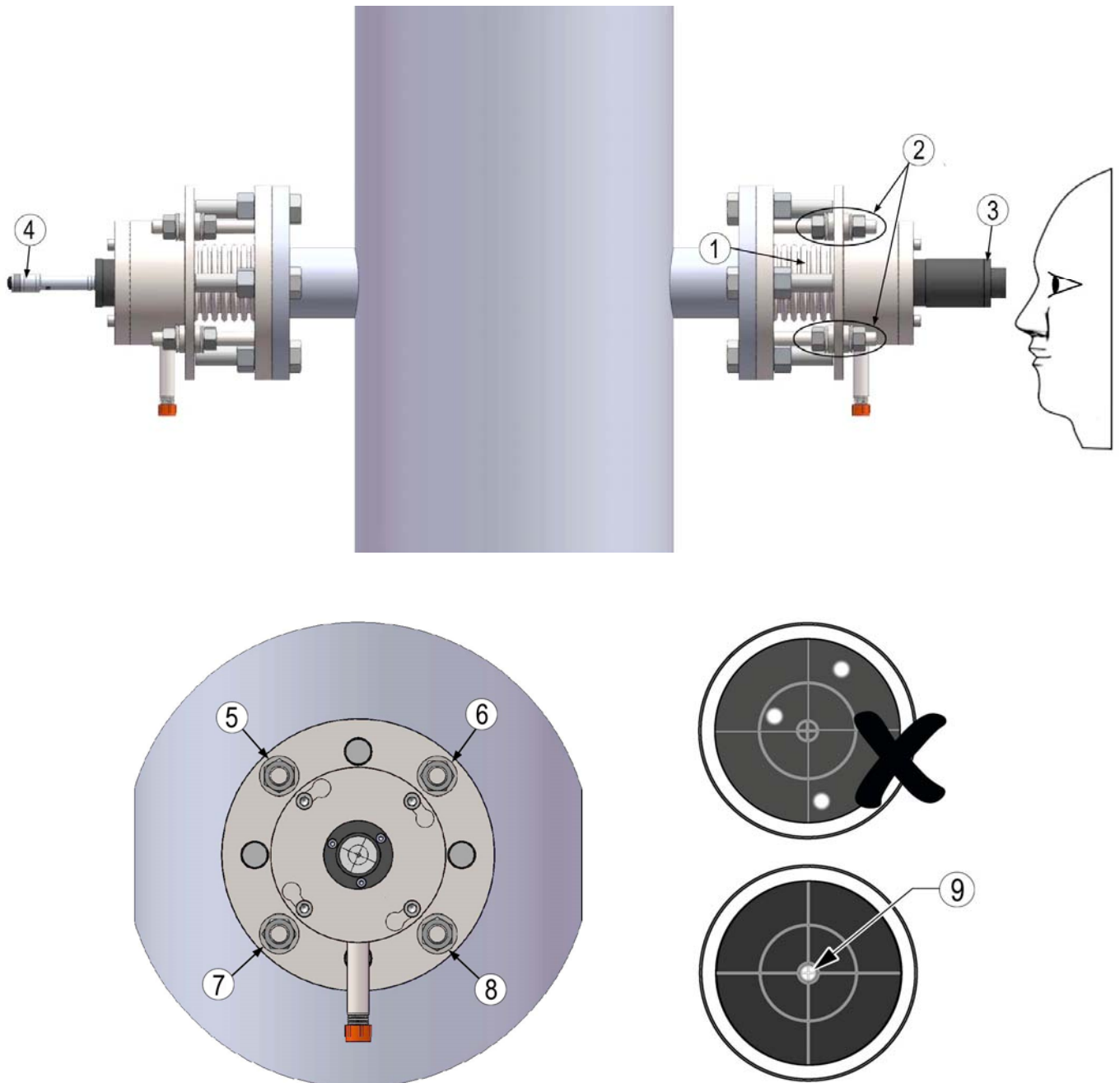
Figure 3-9 – Overview of the alignment tool mounted on the duct

Carrying out the preliminary alignment:

This operation should be carried out very precisely. It consists in aligning the light beam from the light source to the center of the aiming device using the alignment nuts of the alignment bolts placed above the alignment bellows (1). There are four alignment bolts, each with two alignment nuts (2).

- Look through the aiming device (3) (receiver side) towards the light source (4).
- Hand-tighten the front nut (5). Fully loosen nuts (6), (7) and (8).
- By hand, move the alignment bellows until the light spot is visible on the aiming device target.
- By hand, retighten the alignment nuts to hold the alignment bellows in position.
- Adjust the alignment nuts (5), (6), (7) and (8) to adjust the aiming device direction.
- Position the light spot on the centre of the aiming device target (9).
- Hand-tighten all the nuts to keep the light spot in place.

The preliminary alignment steps are completed. The final alignment step is alignment using Web Server.



(1) Alignment bellows, (2) alignment nuts, (3) aiming device, (4) light source, (5), (6), (7), (8) nuts, (9) correct target of the aiming device.

Figure 3–10 – Preliminary alignment procedure

3.1.3.4. Mounting the analyzer (transmitter and receiver)

NOTE : Before mounting the analyzer on the alignment flanges above the bellows, it is necessary to remove the alignment tools.

Mounting the Transmitter:

The transmitter should be mounted **on the torch light side**.

- Turn off the light source (1). With the 5 mm Allen Key, loosen the four nuts of the bolts (5) retaining the light source disc on the transmitter alignment flange (3), and remove the light source disc. (See Figure 3–8).
- Insert the drill holes of the transmitter in the bolt of the flange. Turn slowly the transmitter toward the narrow side of the drill holes (1) until they lock. (See Figure 3–11). Tighten the nuts with the 5 mm Allen Key.

Mounting the Receiver:

The receiver should be mounted **on the aiming device side**.

- With the 5 mm Allen Key, loosen the four nuts of the bolts (8) retaining the aiming device disc on the transmitter alignment flange (6), and remove the aiming device. (See Figure 3–8).
- Insert the drill holes of the receiver in the bolt of the flange. Turn slowly the receiver toward the narrow side of the drill holes until they lock. Tighten the nuts the 5 mm Allen Key.



(1) narrow sides of the drill holes

Figure 3–11 – Mounting the analyzer (transmitter and receiver)

3.1.3.5. Electrical connecting the analyzer

When preliminary alignment is complete and the Transmitter and Receiver have been mounted, begin working with electrical connections.

The cable specifications are indicated in section 1.2.1.4.

Cables must be connected to connection devices with protection mode listed in EN/IEC 60079-0.



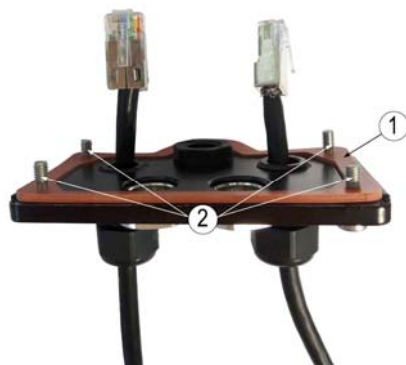
Power off the power supply before working with electrical cables and connections.

The analyzer is supplied with fitting plates mounted on its lower panels. This operation consists in removing the fitting plates located on the lower panels and replacing them by cable gland plates on which are assembled the cables to be connected to the analyzer.

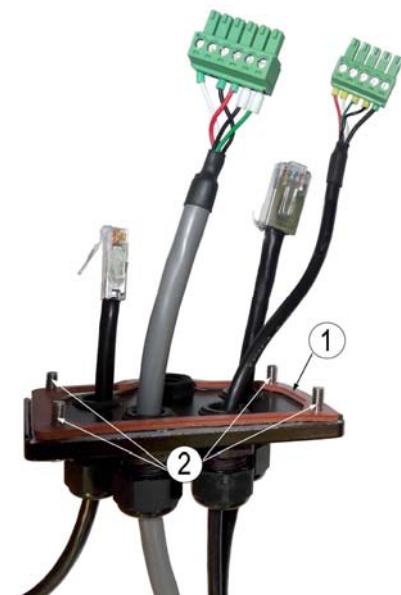
- Remove cable gland plate assembly from the packaging: the cable gland plates are equipped with a sealing gasket (1) and four captive screws (2).
- Check that the gland plate sealing gasket is clean and free from dust. Clean, if necessary, with a non-abrasive cloth.



- The cable gland plate with **two cables** should be connected to the **transmitter**,
- The cable gland plate with **four cables** should be connected to the **receiver**.



Transmitter cable gland plate



Receiver cable gland plate

Figure 3–12 – Cable gland plate of transmitter and receiver

Transmitter electrical connection:

- With the 3 mm Allen Key, unscrew and remove the 4 screws (1) keeping in place the transmitter fitting plate, and remove the plate. Loosen the four screws (2) holding closed the transmitter door and open the door.
- Position the transmitter cable gland plate so that the screws (3) fit in the holes (4) of the lower panel. Tighten the screws.
- Connect the connectors (5) and (6) as shown on the figure: (5) is the master-slave connector and (6) is the ethernet connector.

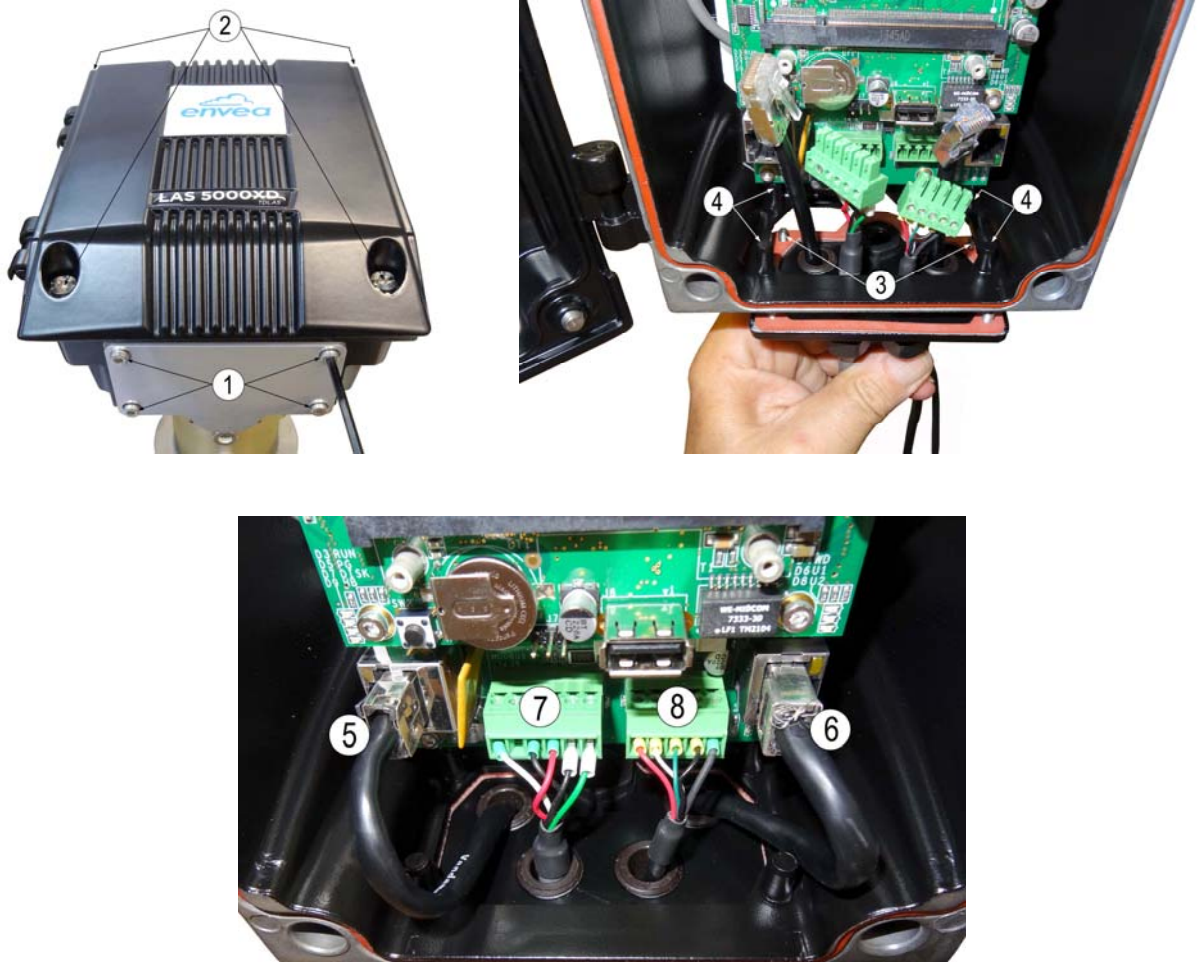


(1) Fitting plate screws, (2) door screws, (3) screws of the cable gland plate, (4) screws hole shown upside down, (5) receiver (master) - transmitter (slave) connector (RJ50), (6) ethernet connector (RJ45).

Figure 3-13 – Transmitter electrical connection

Receiver electrical connection:

- With the 3 mm Allen Key, unscrew and remove the 4 screws (1) keeping in place the receiver fitting plate, and remove the plate. Loosen the four screws (2) holding closed the receiver door and open the door.
- Position the receiver cable gland plate so that the screws (3) fit in the holes (4) of the lower panel. Tighten the screws.
- Connect the connectors (5), (6), (7) and (8) as shown on the figure : (5) is the master-slave connector, (6) is the ethernet connector, (7) is the 24V power supply + Modbus, (8) is the USB Estel connector.

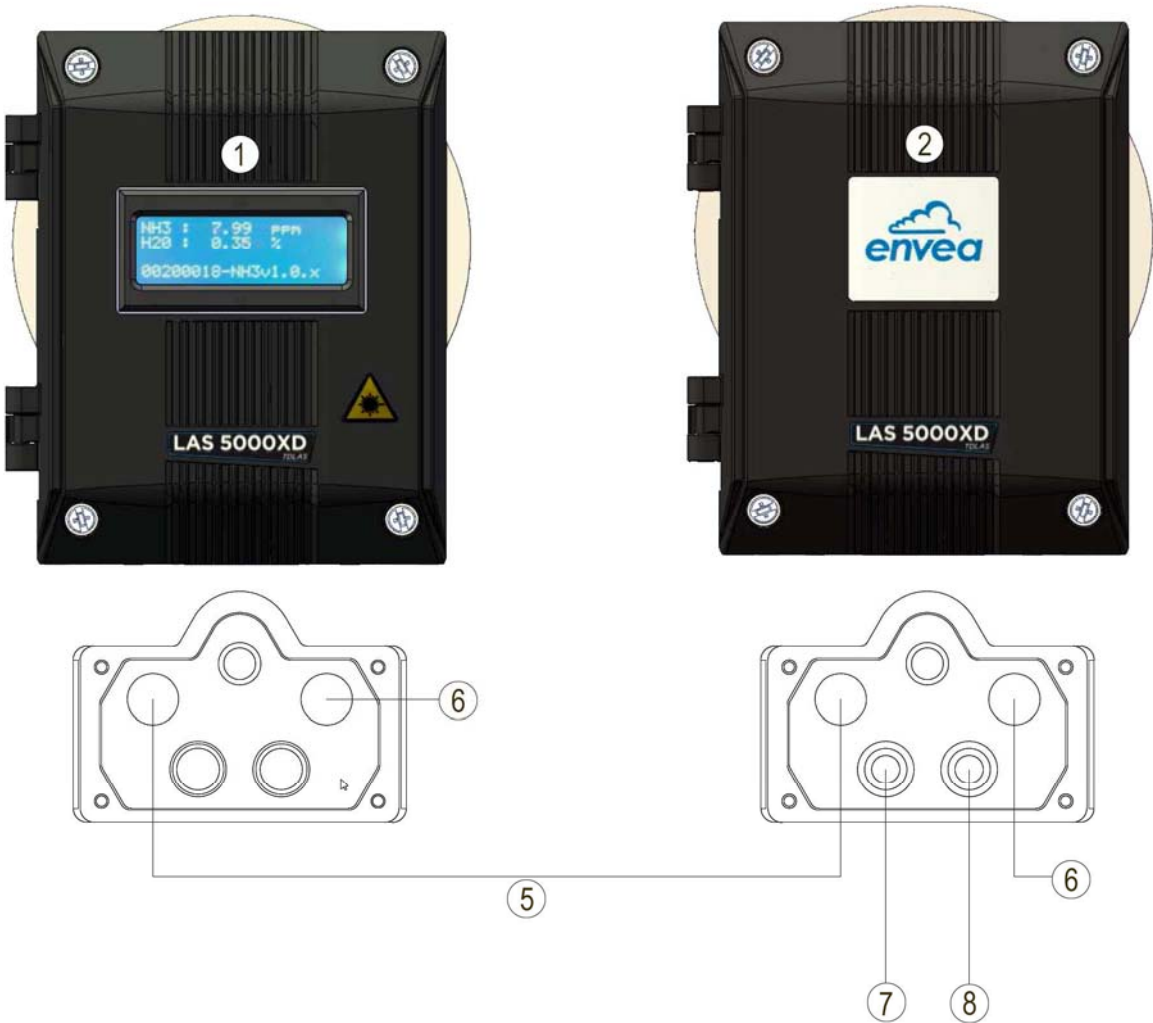


(1) Fitting plate screws, (2) door screws, (3) screws of the cable gland plate, (4) screws hole shown upside down, (5) receiver (master) - transmitter (slave) connector (RJ50), (6) ethernet connector (RJ45), (7) 24V power supply + Modbus RTU connector, (8) USB Estel connector.

Figure 3-14 – Receiver electrical connection

Transmitter (slave)-Receiver (master) electrical connections

The overview diagram shown below enables the user to have a global view of the transmitter (slave) receiver (master) electrical connections.



(1) Transmitter (slave), (2) Receiver (master), (5) Receiver (master) - transmitter (slave) connection (RJ50), (6) Ethernet connection (RJ45), (7) 24V power supply + Modbus RTU connection, (8) USB Estel (AI/AO) connection.

Figure 3–15 – Overview diagram of transmitter (slave) receiver (master) electrical connections

Ethernet connector (RJ45) is used for ethernet switch/laptop (web interface)

Wires of 24V power supply + modbus connector (7)					
1	2	3	4	5	6
White	Not connected	Black	Red	Black	Green
PE > 1		GND > 2	24VDC > 4	ModBus RTU (-) > 6	ModBus RTU (+) > 7

Modbus RTU = RS485

Analyzer connection to the 24V power supply:**WARNING!****QUALIFIED PERSONNEL ONLY!**

Electrical work must be carried out only by suitably trained and qualified personnel.

There are two possibilities for connecting the analyzer to the 24V power supply:

- Connect the Power / Modbus RTU cable to a 24V power supply (not provided), according to the information on the cable labels.

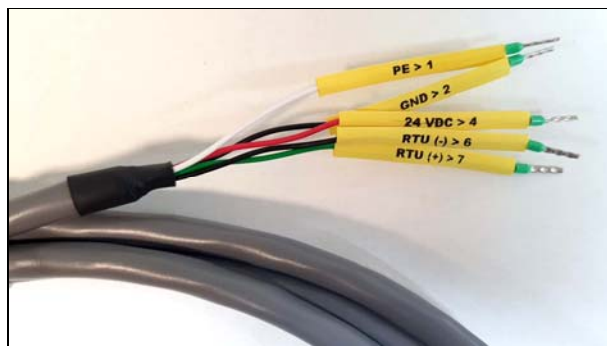
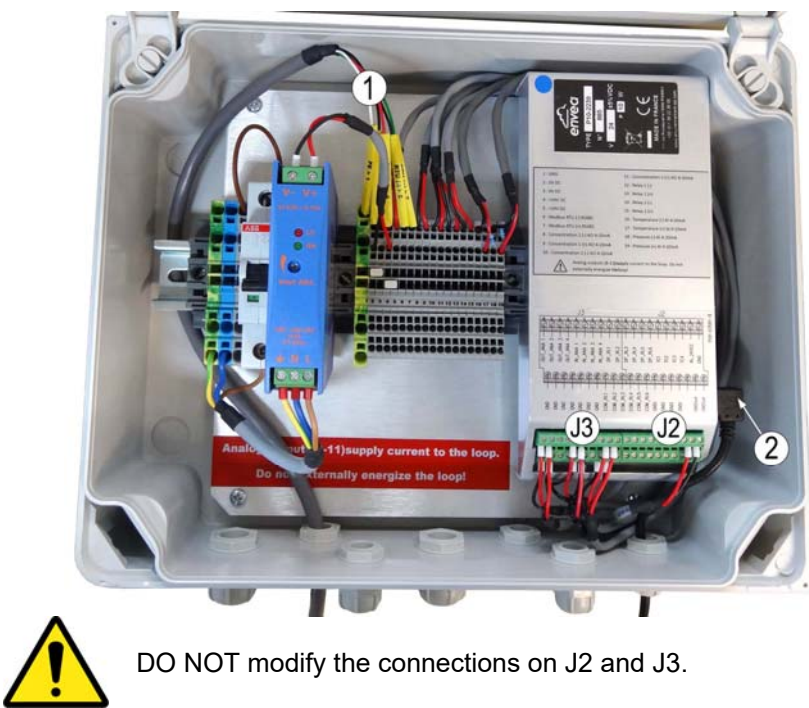


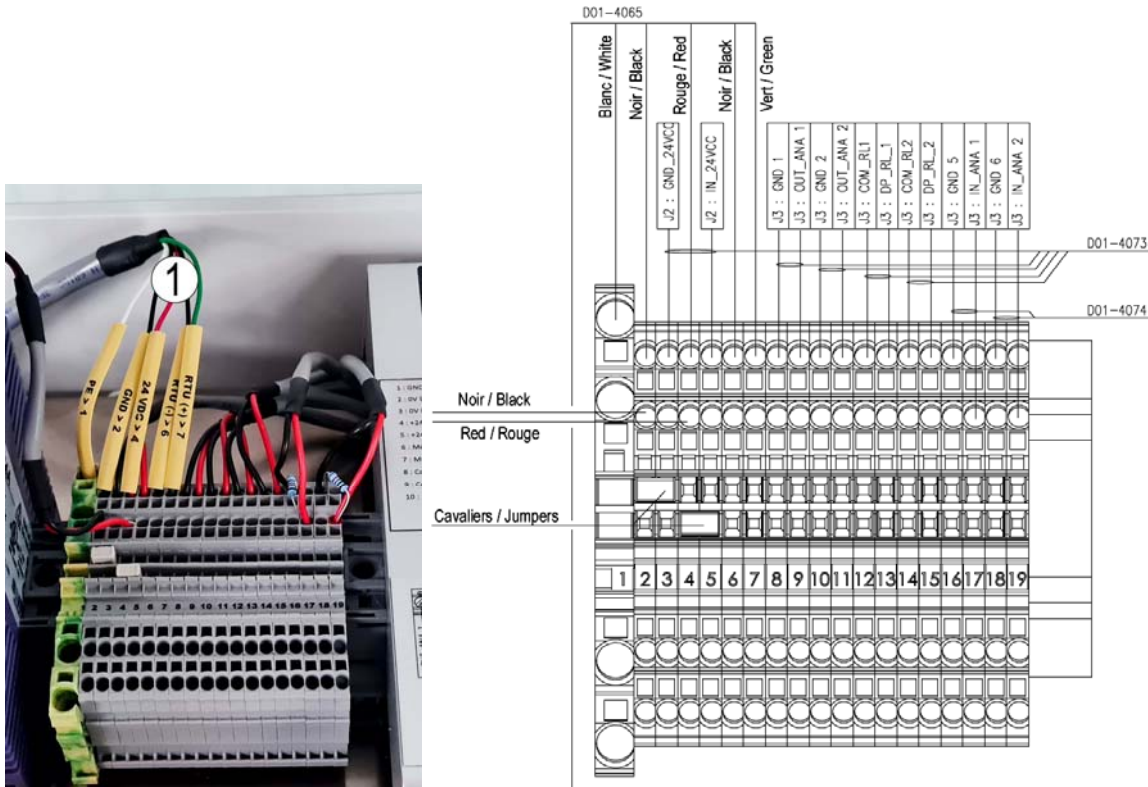
Figure 3–16 – Power / Modbus RTU cable

or

- Connect the Power / Modbus RTU cable (1) and the USB ESTEL cable (2) to the optional Junction box following the indications of Figure 3–17.



DO NOT modify the connections on J2 and J3.



(1) Power / Modbus RTU cable, (2) USB ESTEL cable, D01-4065: Modbus cable, D01-4073: relay, concentration, power supply cable, D01-4074 : pressure, temperature cable

Figure 3-17 – Junction box connections

 Analog outputs (8-11) supply current to the loop. DO NOT externally energize the loop.

3.1.3.6. Electrical powering on



Power on the LAS 5000XD after all previous steps in this chapter have been completed and it is safe to do so.

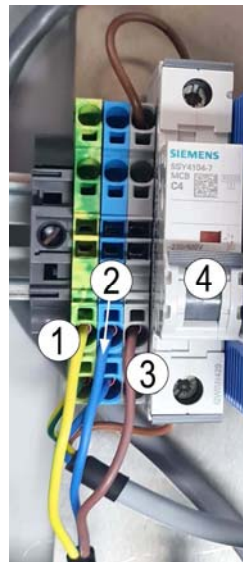
WARNING!**INVISIBLE LASER RADIATION!**

The LAS 5000XD uses lasers emitting light in the near-infrared region of the spectrum. When installed on a duct or stack, the LAS 5000XD is classified as a Class 3R et 3B laser product.

The analyzer supports 24V only

Without the junction box, connect the 24V power supply of the analyzer to the mains of the site.

With the junction box (refer to section 1.1.4.1 for junction box description), connect the (1) ground, (2) neutral, (3) phase wires on the mains cable to the terminal block as indicated at Figure 3–18, and then switch on the circuit breaker (4).



- (1) Ground
- (2) Neutral
- (3) Phase
- (4) Circuit breaker

Figure 3–18 – Powering on the analyzer with the Junction box



The analyzer will start up **2min30 secs** after powering on. Please, wait for it.

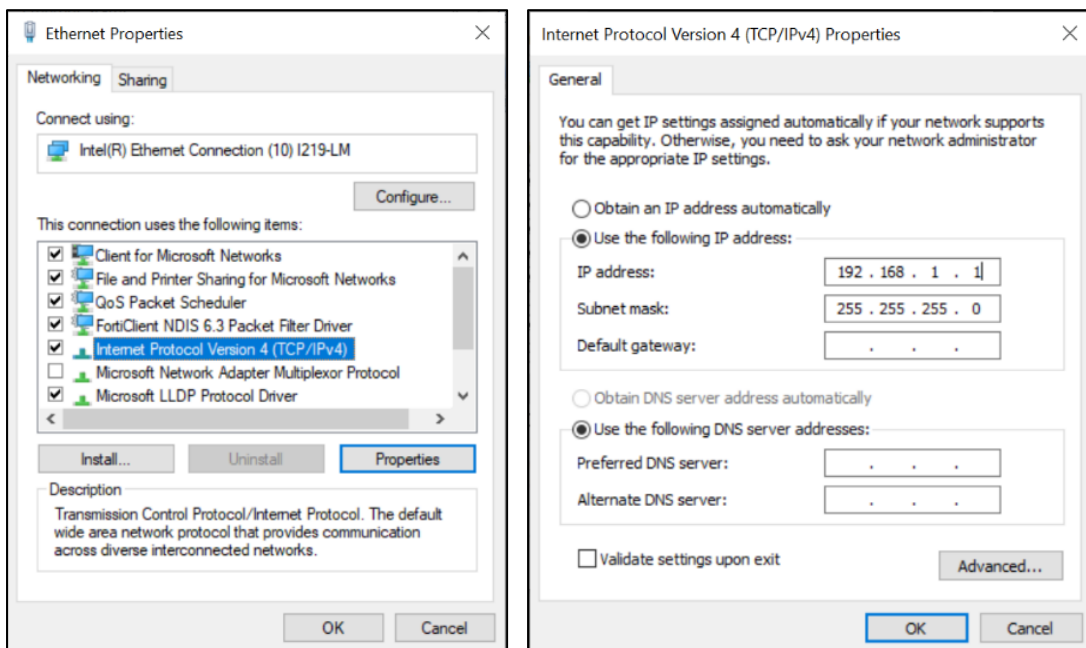
At startup, the screen display shows the following information:

Gas and Light Transmission Values	Modbus and Software version
<pre>NH3 0.46 PPM H2O 0.00 % Trans: 100 % OK</pre>	<pre>Modbus RTU ID : 1 FirmwareM: 1.1.c_rc2 FirmwareS: 1.1.b_rc1 000-LAS5000XD-NH3</pre>
IP Addresses and Mask	Modbus
<pre>IPM: 192.168.1.5 IPS: 192.168.1.6 000-LAS5000XD-NH3</pre>	<pre>Baud : 19200 Bits : 8 Stop : 1 Parity : none</pre>

Connect the RJ45 cable of the receiver (master) to the ethernet network in order to access the web interface with your laptop. The default addresses are:



If the network is not compatible, the user should set the IP address of his computer to static.



Then, input 192.168.1.5:5000 in the address bar to connect to the WebServer. The home page is displayed as follows:

The screenshot displays the ENVEA LAS 5000XD web interface. At the top, the browser address bar shows '192.168.1.5:5000'. The main header shows 'NH3 0.00 ppm' and 'H2O 0.79 %'. Below the header is a navigation bar with icons for home, settings, graphs, and data. A 3D model of the analyzer is shown on the left. On the right, there are two tables: 'Gas 1' for NH3 and 'Gas 2' for H2O, both showing average, k-factor, and offset. Below these are 'Outside Path' settings for concentration, pathlength, temperature, and pressure. A 'Laser Tool' window is open at the top left.

Parameter	Value	Unit
Pathlength	39.0	cm
Temperature	24.0	°C
Pressure	1.0132	bar
Master Internal T.	36.44	°C
Slave Internal T.	34.06	°C
Laser T.	32.358	°C
Optical Power	27446	
Ref. Power	26728	
Transmission	103	
Gain	1	
Auto Gain	OFF	

Parameter	Value	Unit
Concentration	0.000	%
Pathlength	39	cm
Temperature	24	°C
Pressure	1.01325	bar

Click on the “Settings” button , then on the “Communication” button to display this sub-menu. Change the IP address of the analyzer with static mode selected, or switch to DHCP mode if DHCP server is available. Then, valid the modifications with the set button .

The screenshot displays the ENVEA LAS 5000XD web interface 'Communication' sub-menu. The main header shows 'NH3 0.05 ppm' and 'H2O 0.64 %'. Below the header is a navigation bar with icons for home, settings, graphs, and data. The 'Communication' button is highlighted. The sub-menu contains two tables: 'Network' and 'Modbus RTU'. The 'Network' table shows Mode (Static), IP (172.16.13.250), Mask (255.255.0.0), and Gateway (172.16.1.230). The 'Modbus RTU' table shows settings for Enable, Baud rate, Stop bits, Parity, Data, Slave ID, Swap words, Polarisation, and Termination.

Parameter	Value	Unit
Mode	Static	
IP	172.16.13.250	
Mask	255.255.0.0	
Gateway	172.16.1.230	

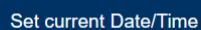
Parameter	Value	Unit
Enable	ON	on/off
Baud rate	19200	
Stop bits	1	
Parity	None	
Data	8	
Slave ID	1	
Swap words	OFF	on/off
Polarisation	ON	on/off
Termination	ON	on/off

NOTE : If the DHCP mode is selected, the IP address of the Master, called IPM, will be displayed on the Transmitter screen every 60 seconds.

3.1.3.7. Date and time updating

If necessary, update the date and time of the analyzer to the current date. Click on the “Information” button  to display the corresponding page, and click on the button  to set the Date and Time fields to the current computer date to which the analyzer is connected. Then, click on  of the Date field and check that the message “Setting acknowledged” appears to be sure the modifications are taken into account. Do the same for the Time field.

Date	18 / 10 / 2022		
Time	17 : 27 : 21		



3.1.3.8. Final alignment using the WebServer

This is the second step of the alignment started in section 3.1.3.3, and have to be done when the duct is running.



Make sure the LAS 5000XD is working on a live process before carrying out the second alignment.

The WebServer final alignment allows to define the Reference Power of the laser beam (i.e. the maximum Optical Power value of the laser) and to set it in the WebServer. The Reference Power corresponds to a 100% transmission.

This alignment consists in adjusting the nuts of the alignment bellow bolts of both Transmitter and Receiver until the WebServer displays the maximum possible reading.



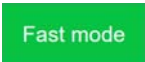
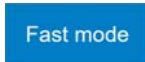
To do that:

- Click on the « Alignment » button  to display the corresponding page and see the Optical Power value (1).
- Begin the alignment with the transmitter side. Click on the Fast mode button  to carry out the alignment settings. This button gets green  meaning that Fast mode is activated. When the maximum value is obtained, fully tighten all the nuts of the alignment bellow bolts in order not to change the position of the Transmitter.
- Perform the same operation on the receiver side in order to still increase the value previously obtained on the transmitter side.

- When the maximum value of the receiver is reached, click on the “Set Trans.” button  to record this value as Reference Power in the corresponding field of the home page which will indicate a 100% Transmission.

Optical Power	27441
Ref. Power	27444
Transmission	100

NOTE : In normal operation, if the Optical Power decreases compared to the Ref. Power, the Transmission will be lower than 100% indicating that the window gets dirty or the analyzer is out of alignment.

Click on  to exit the Alignment mode. This button gets blue  meaning that mode is disabled.

At any time, observe that the message “Setting acknowledged” appears to be sure the settings are taken into account.



Now, the analyzer is ready to operate.

3.2. WEBSERVER DESCRIPTION

The LAS 5000XD uses an HMI (Human Machine Interface) software, named WebServer, for a number of tasks, including aligning the Transmitter and Receiver, setting up the LAS 5000XD, logging data to a file, and diagnostics.

The WebServer home page is shown below. The instant value of the concentration measures in the selected unit is displayed at the top of all the pages, and the user just needs to click on the browsing button to display the corresponding page.

The buttons are white on a blue background which turns grey when selected.

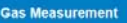


The functionalities of the browsing buttons and their corresponding pages are as follow:

Button	Page name	Function (used to)
	Home page	This page displays the current operating and measurement parameter values of the analyzer. As they are on grey background, these values cannot be modified by the user.
	Alignment	This page displays the current value of the laser optical power and enables to carry out the second alignment.
	Record	By default, this page displays in live the current concentration values of the gases being sampled.
	Diagnostics	By default, this page displays in continuous the current concentration values of the gases being sampled, the optical power of the laser, the pressure and the temperature in the stack between the two flanges.
	Spectra	This page gives a plot of the laser radiation spectrum transmitted through the gas and fitted by a mathematical model. It provides also the gas concentration independently of the Laser power.
	Configuration menu	This page gives access to configurable sub-menus: Gas measurement, Communication, 4-20 mA, Audit Cell mode, Software update, Engineer.

Button	Page name	Function (used to)
	Technical manual	This button allows the user to load the technical manual of the LAS 5000XD in the download file.
	Information panel	It gives the information necessary to identify the analyzer and to communicate with.

The Configuration button  gives access to the following pages :

Button	Page name	Function (used to)
	Gas measurement sub-menu	This sub-menu allows to configure the instrument, temperature, pressure and outside path parameters involved in the gas measurement, and the settings of the measured gases.
	Communication sub-menu	This sub-menu allows to manage the network and Modbus RTU communications.
	4-20 mA sub-menu	This sub-menu allows to configure the analog inputs / outputs.
	Audit Cell mode sub-menu	The audit cell is an optional device allowing to easily check the accuracy of the measurements given by the analyzer. It is mounted outside the stack and is completely independent from its fluid circuit.
	Software Update sub-menu	This sub-menu allows to update this WebServer software. The software update file should be requested from ts@envea.global. It is named "update.sh".
	Engineer sub-menu	This sub-menu access is for ENVEA Service Engineers only.

Parameters modifications:

Only the fields on white background can be modified, the fields on grey background cannot be modified.

When the user modifies a field, he should valid the modification by clicking on  or  and

watch for the message  appear-disappear in the top right corner of the page to be sure the modification has correctly been taken into account. As it is an instant message, it disappears as quickly as it has appeared.

Analyzer status indication :

Current status of the analyzer is indicated by a spot displayed at the top right corner of each page. This spot is green  when the analyzer works normally (no alarm present). It turns red when an alarm is triggered. Then, the user has to click on this red point to display the current alarms and take the suitable actions to solve them.

3.2.1. HOME PAGE

The user displays this page by clicking on



which becomes



This page displays the current operating and measurement parameter values of the analyzer. As they are on grey background, these values cannot be modified by the user.

The parameters are the followings:

- PATHLENGTH field: displays the length of the optical path between the two flanges of the stack.
- TEMPERATURE field: indicates the temperature value in the stack between the two flanges.
- PRESSURE field: indicates the pressure value in the stack between the two flanges.

A “Static” or “Dynamic” message is displayed next to each temperature and pressure field meaning that the operating mode can be static or dynamic:

- Static: there is no reference value and no sensor. Consequently, the user fixes the temperature and pressure values by himself.
- Dynamic: a sensor gives these measurements via the analog input/output ESTEL board of the Junction Box connected to the analyzer. Consequently, no action is required from the user.

NOTE : These two modes are set in the GAS MEASUREMENT page  of the CONFIGURATION menu .

- MASTER INTERNAL T. field: indicates the internal temperature of the master, measured by the sensor present on the I/O board mounted inside the receiver.
- SLAVE INTERNAL T. field: indicates the internal temperature of the slave, measured by the sensor present on the I/O board mounted inside the transmitter.
- LASER T. field: gives the laser temperature measured.
- OPTICAL POWER field: indicates the current laser optical power detected by the receiver.
- REF. POWER field: indicates the reference power value for the considered laser detected by the receiver.
- TRANSMISSION field: it is the transmission capability of the laser. When Optical Power equals Reference Power, transmission is 100%. When transmission decreases, it indicates that the window has got dirty and should be cleaned, or that the alignment should be checked.
- GAIN field: it is the gain (amplification) applied at signal acquisition level, i.e. the optical power detected at receiver level. When Gain is 1, no multiplying coefficient is applied.
- AUTO GAIN field: When ON, the gain is applied automatically. When OFF, the gain is disabled.

GAS 1 and GAS 2 sections:

- NAME field: it is the name of the gas measured by the analyzer. In the present example, two gases are measured, NH3 and H2O.
- AVERAGE field: the concentration value displayed is an average of instantaneous measurements. This field indicates the number of successive samples averaged to obtain a measurement.
- K-FACTOR field: it is the span coefficient of the considered gas. The user can set it in the Configuration menu, Gas Measurement page.
- OFFSET field: it indicates the offset value applied to the considered measurements.

OUTSIDE PATH section :

Since the gas is only measured along the laser path length through the stack, and the total laser path length is longer because of the flanges, the transmitter and the receiver, it is necessary to take into account the laser path length of the flanges, the transmitter and the receiver on both sides and remove it for the concentration calculation.

Consequently, this section gives all the necessary information about the optical path due to flanges, transmitter and receiver which will be used in calculations.

- CONCENTRATION field: it is the concentration value (in %) to be removed from the measurement because of the flange path length.
- PATHLENGTH field: gives the optical path length of the flanges only.
- TEMPERATURE and PRESSURE fields: indicates the temperature and pressure values at flange level. They will be used in calculations.



NH3 0.00 ppm **H2O 0.79 %**

18/10/2022 13:11:45

Pathlength	39.0	cm
Temperature	24.0	°C
Pressure	1.0132	bar

Master Internal T. 36.44 °C
Slave Internal T. 34.06 °C
Laser T. 32.358 °C

Optical Power	27446
Ref. Power	26728
Transmission	103

Gain	1
Auto Gain	OFF

Gas 1

Name	NH3
Average	5
k-factor	1.000
Offset	0.000

Gas 2

Name	H2O
Average	5
k-factor	1.000
Offset	0.000

Outside Path

Concentration	0.000	%
Pathlength	39	cm
Temperature	24	°C
Pressure	1.01325	bar

3.2.2. ALIGNMENT PAGE

The user displays this page by clicking on



which becomes



This page displays the current value of the laser optical power (1) and enables to carry out the second alignment.

The curves represent the Optical power detected by the receiver as a function of the laser current stated in point number (Data points). They allow to visualize and check the optical power linearity.

The Fast Mode button **Fast mode** is used to switch the analyzer in Alignment mode. This special mode stops the measurements and enables to find the maximum Optical power of the laser. As explained previously, when the maximum value of the receiver is reached, it should be recorded with the Set. Trans button **Set Trans.**. It is the Reference Power value.

Click on **Fast mode** to exit Fast mode. This button turns blue again meaning that Fast mode is disabled.

NOTE : Since the Fast Mode stops the measurement when the process is running, it is recommended to carry out the alignment when the process is stopped.



3.2.3. RECORDS PAGE

The user displays this page by clicking on



which becomes



By default, this page displays in continuous the current concentration values of the gases being sampled.

During data acquisition, the analyzer stores the data in four different databases: “Seconds”, “Hours”, “Days” and “Weeks”. Thanks to the buttons available at the upper part of the page, the user can display the stored data from the different databases and for different time intervals:



: the “Month” button displays the “Hours” data for a month.



: the “Week” button displays the “Minutes” data for a week.



: the “Day” button displays the “Seconds” data for a day.

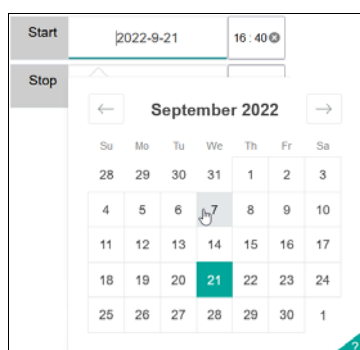


: the “Hour” button displays the data recorded in the “Seconds” database for an hour.



: the “Live” button displays the current measurements.

When the user clicks on the “Month” button, such will automatically fill the “Start” and “Stop” input fields from the current date (stop) to the same date of the previous month (start). This default setting can be modified as desired by clicking on the date input fields to open a selection pop-up for selecting another date and closing the pop-up.



In the same way, the “Week” button allows to reset the date and hour input field within one week, the “Day” button within one day and the “Hour” button within one hour.

The user clicks on the Plot button

Plot

to display the selected data as graphs.

He clicks on the Download Current View button

Download Current View

to export the selected data in a csv file and visualize them.

Additionally, this graph offers associated functionalities as zoom by selecting with the mouse pointer a specific area in the graph, and display date, hour and concentrations on a specific point.

The user zooms out by left double-clicking anywhere in the graph.



3.2.4. DIAGNOSTICS PAGE

The user displays this page by clicking on



which becomes



By default, this page displays in continuous the current concentration values of the gases being sampled, the optical power of the laser, the pressure and the temperature in the stack between the two flanges.

During data acquisition, the analyzer stores the data in four different databases: "Seconds", "Hours", "Days" and "Weeks". Thanks to the buttons available at the upper part of the page, the user can display the stored data from the different databases and for different time intervals: the button functionalities of this page are the same as those of the Record page.

The legends (1) at the top of the screen are active. By clicking on them, the user enables/disables the display of the corresponding curve. The information given in the graph by the mouse pointer (2) allows a visual comparison of the measurements at a given time and facilitates the diagnosis.



3.2.5. SPECTRA

The user displays this page by clicking on



which becomes



This page gives a plot of the laser radiation spectrum emitted by the transmitter (Transmission) and the measurement acquired at the receiver (Measurement) as a function of the relative wave number.

The Transmission curve is plotted from raw (not fitted) data, the Measurement curve is fitted according to the average values.

The legends (1) at the top left of the screen are active. By clicking on them, the user enables/disables the display of the corresponding curve.

The information given in the graph by the mouse pointer (2) allows a visual comparison of the measurements at a given time and facilitates the diagnosis.

The download data button

Download data

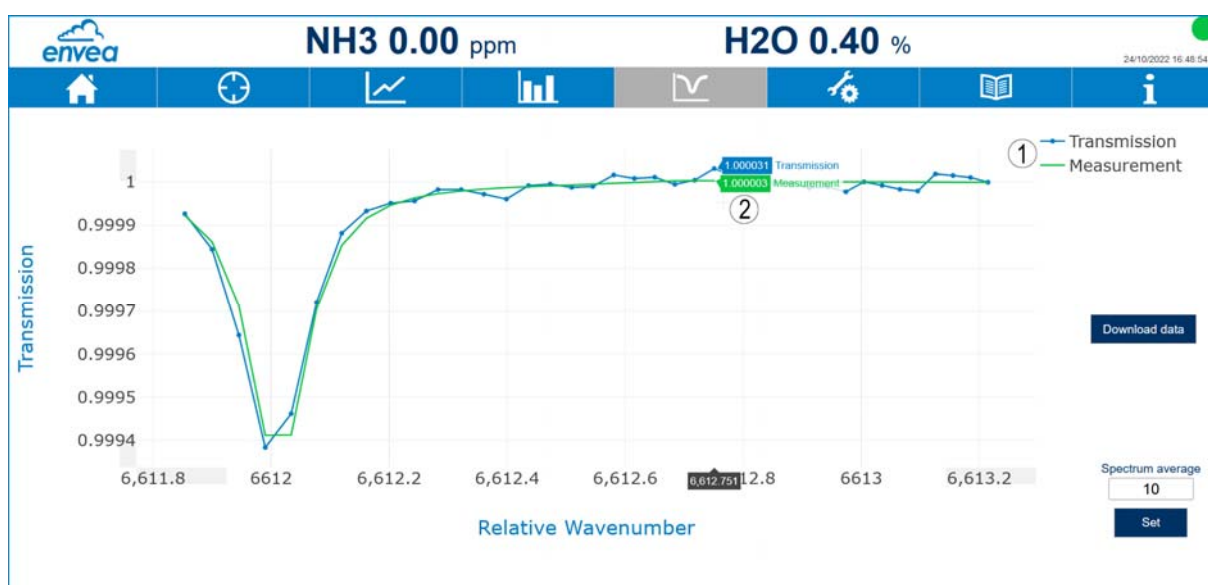
enables to download the measurement data in a CVS file.

The Spectrum average setting box allows to modify the measurement number from 1 to 10 to be averaged to fit the measurement.

Consequently, the Set button

Set

allows to fit again the Measurement curve with this new average setting.



The relative Wavenumbers are stated in cm^{-1} .

3.2.6. CONFIGURATION

The user displays this page by clicking on  which becomes  and gives access to configurable sub-menus: Gas measurement, Communication, 4-20 mA, Audit Cell mode, Software update, Engineer.

3.2.6.1. Gas Measurement

The user displays this sub-menu by clicking on  which becomes . It allows to configure the instrument, temperature, pressure and outside path parameters involved in the gas measurement, and the settings of the measured gases.

INSTRUMENT section: it allows to set the specific parameters of the analyzer.

- **PATHLENGTH** field: resets the length of the optical path between the two flanges of the stack.
- **AUTO GAIN** field: enables (ON) /disables (OFF) the automatic gain mode of the analyzer. When ON, the gain is applied automatically.
- **FIXED GAIN** field: when **AUTO GAIN** field is OFF, it allows to set the gain (amplification) value applied at signal acquisition level, i.e. the optical power detected at receiver level. When Gain is 1, no multiplying coefficient is applied.
- **NEGATIVE VALUES** field: negative values are allowed when this field is ON, and not allowed when it is OFF.

TEMPERATURE section: it allows to set the parameters applied to the temperature measured in the stack between the two flanges.

- **GAS TEMP. MODE** field: allows to select the “Static” or “Dynamic” mode of the gas temperature measurement. When Static is selected, the user fixes the temperature value by himself in the **GAS TEMPERATURE** field. When Dynamic is selected, this setting is not allowed.
- **GAS TEMPERATURE** field: sets the gas temperature value when the Static mode is selected.

PRESSURE section: it allows to set the parameters applied to the pressure measured in the stack between the two flanges.

- **GAS PR. MODE** field: sets the “Static” or “Dynamic” mode of the gas pressure measurement. When static is selected, the user fixes the pressure value by himself in the **GAS PR.** field. When dynamic is selected, this setting is not allowed.
- **GAS PR. Field** : sets the pressure value when the static mode is selected.

OUTSIDE PATH section: as mentioned at the homepage description, the gas is only measured along the laser path length through the stack. Since the total laser path length is longer because of the flanges, the transmitter and the receiver, it is necessary to take into account the laser path length of the flanges, the transmitter and the receiver on both sides and remove it from the concentration calculation.

Consequently, in this section the user can set all the necessary information about the optical path due to flanges, transmitter and receiver which will be used in calculations.

- **CONCENTRATION** field: sets the concentration value (in % or RH) to be removed from the measurement because of the flange, transmitter and receiver path length.
- **PATHLENGTH** field: sets the optical path length due to the flanges, transmitter and receiver only.
- **TEMPERATURE** and **PRESSURE** fields: sets the temperature and pressure values at flange, transmitter and receiver level.



These parameters are set in factory before delivery and it is strongly advised not to modify them. If the pathlength is modified, that will modify completely the metrology.

NH3 (gas 1) and H2O (gas 2) sections : it allows to manage the parameters of the gas measurement.

- AVERAGE field: the concentration value displayed is an average of instantaneous measurements. This field allows to set the number of successive samples averaged to obtain a measurement.
- K-FACTOR field: sets the span coefficient of the gas to be measured. (It is not recommended to modify this value).
- OFFSET field: sets the offset value applied to the gas to be measured. (It is not recommended to modify this value).
- UNITS field: allows to select the gas measurement unit.

envea		NH3 0.00 ppm		H2O 0.41 %		25/10/2022 11:29:31	
Gas Measurement		Communication		4-20 mA		Audit Cell mode	
Software Update		Engineer					
Instrument				Outside Path		NH3	
Pathlength	39.0	cm	✓	Concentration	0.000	%	✓
Auto Gain	OFF	on/off	✓	Pathlength	39	cm	✓
Fixed Gain	1		✓	Pressure	1.01325	bar(a)	✓
Negative values	OFF	on/off	✓	Temperature	24	°C	✓
Temperature						H2O	
Gas Temp. Mode	Static		✓			Average	5
Gas Temperature	24.0	°C	✓			K-factor	1.000
Pressure						Offset	0.000
Gas Pr. Mode	Static		✓			Units	ppm
Gas Pr.	1.0132	bar(a)	✓				
						Average	5
						K-factor	1.000
						Offset	0.000
						Units	%

3.2.6.2. Communication

The user displays this sub-menu by clicking on **Communication** which becomes **Communication**. It allows to manage the network and Modbus RTU communications.

NETWORK section: it allows the automatic network configuration. The MODE field allows to select the STATIC or DHCP mode:

- When DHCP is selected, it allows connecting the analyzer to a TCP/IP network address managed by a DHCP server. Consequently, the user cannot modify the IP address, the mask and the gateway fields.
- When STATIC is selected, the ADDRESS, MASK and GATEWAY fields can be modified. Thus, the user defines the TCP / IP address, gateway and mask required for connection.

The user should click on the set button **Set** for the modifications to be taken into account.

MODBUS RTU section: it allows to manage the MODBUS communication on RS.

- ENABLE field: allows to activate (ON)/ deactivate (OFF) the MODBUS.
- BAUD RATE field: allows to select signal speed among 2400, 4800, 9600, 19200, 38400, 57600, 115200 bds.
- STOP BITS field: allows to select the stop bits. The frame transmitted has only one or two stop bits.
- PARITY field: this field should be set on "NONE" since the client-server parameters have to be the same for correct communication.
- DATA field: it indicates the data format.
- SLAVE ID field: this is the analyzer identifier. For correct communication, all the Modbus frames to be sent should have an identifier equivalent to the Slave ID.
- SWAP WORDS field: when ON, high/low byte exchange is allowed. When OFF, high/low byte exchange is not allowed.
- POLARISATION field: when ON, the polarization resistance is activated. When OFF, it is deactivated.
- TERMINATION field: when ON, the termination resistance is activated. When OFF, it is deactivated.

The screenshot displays the ENVEA LAS 5000XD interface. At the top, it shows gas measurements: NH3 0.00 ppm and H2O 0.51 %. The 'Communication' menu is selected, showing two main sections: Network and Modbus RTU.

Network Configuration:

Field	Value
Mode	Static
IP	172.16.13.250
Mask	255.255.0.0
Gateway	172.16.1.230

A 'Set' button is located below the Network configuration fields.

Modbus RTU Configuration:

Field	Value	Unit/Status
Enable	ON	on/off [power icon]
Baud rate	2400	[checkmark]
Stop bits	1	[checkmark]
Parity	None	[checkmark]
Data	8	[checkmark]
Slave ID	1	[checkmark]
Swap words	OFF	on/off [power icon]
Polarisation	ON	on/off [power icon]
Termination	ON	on/off [power icon]

3.2.6.3. 4-20 mA and analog input/output configuration

The user displays this sub-menu by clicking on **4-20 mA** which becomes **4-20 mA**. It allows to configure the analog inputs / outputs.

ANALOG INPUT RANGE POINT section: it allows to configure the analog inputs according to the specific range of the temperature and pressure sensors.

- **AI RANGE TEMPERATURE** field: since the range of the temperature sensor is 0-200°C, the minimum value of the Analog Input range (4 mA) will be configured to 0°C, and the maximum value of the Analog Input range (20 mA) will be configured to 200°C.
- **AI RANGE PRESSURE** field: since the range of the pressure sensor is 0-2.5000 bars, the minimum value of the Analog Input range (4 mA) will be configured to 0 bars, and the maximum value of the Analog Input range (20 mA) will be configured to 2.5000 bars.

ANALOG OUTPUT 1 and 2 sections : they allow to configure the analog outputs if needed, for instance when the ESTEL board of the Junction Box is changed.

By default, the Analog Output 1 is assigned to the concentration measurement of gas 1 (NH₃ in the present example), and the Analog Output 2 is assigned to the concentration measurement of gas 2 (H₂O in the present example). These settings are carried out in factory before delivery.

- **RANGE** fields : they display the maximum value of the gas concentration range. Consequently, when the corresponding analog output is set, the current value of 20 mA is applied to this maximum value. In the same way, the current value of 4 mA is applied to 0 (minimum value of the gas concentration range).
- **TEST MODE OFF/ON** fields: used to activate/ deactivate the test mode. When ON, the specific fields and appropriate buttons are displayed enabling the user to carry out the analog output settings.
- **TEST VALUE** fields: the user inputs the value of the analog output, and valid with **Set Min.** for the minimum value, and with **Set Max.** for the maximum value.
- **MIN/MAX DAC VALUE** fields: enable to adjust the value read on the multimeter with the analog output value of the analyzer. The DAC values are stated in number of points and ranged from 0 to 4000 pts.

The screenshot shows the ENVEA LAS 5000XD web interface. At the top, it displays the ENVEA logo and gas concentrations: NH₃ 0.00 ppm and H₂O 0.41 %. Below this is a navigation bar with icons for Home, Settings, Graphs, Tables, Audit Cell mode, Software Update, and Engineer. The '4-20 mA' menu is selected, showing the 'Analog Input range point' section with fields for AI Range Temperature (200.0 °C) and AI Range Pressure (2.5000 bar(a)). To the right, the 'Analog Output 1' and 'Analog Output 2' sections are visible. Analog Output 1 has Range (50) and Test mode (OFF). Analog Output 2 has Range (50), Test mode (ON), Test Value (20 mA), Set Min, Set Max, Min DAC value (50), and Max DAC value (3500).

ANALOG INPUT/OUTPUT CONFIGURATION:

The analog inputs are assigned to the temperature and pressure measurement. They are configured in factory before delivery, and they shouldn't be modified.

3.2.6.4. Audit Cell Mode

The user displays this sub-menu by clicking on  which becomes . The audit cell is an optional device allowing to easily check the accuracy of the measurements given by the analyzer. It is mounted outside the stack and is completely independent from its fluid circuit.

INSTALLATION section: it allows to set the specific parameters of the Audit Cell when it is installed.

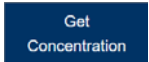
- PATHLENGTH field: resets the length of the optical path of the Audit Cell between the two flanges.
- GAS TEMP. MODE field: allows to select the “Static” or “Dynamic” mode of the span gas temperature inside the Audit Cell. When Static is selected, the user fixes the temperature value by himself in the GAS TEMPERATURE field. When Dynamic is selected, this setting is not allowed.
- GAS TEMPERATURE field: sets the span gas temperature value inside the Audit Cell when the Static mode is selected.
- GAS PR. MODE field: allows to select the “Static” or “Dynamic” mode of the span gas pressure measurement inside the Audit Cell. When Static is selected, the user fixes the pressure value by himself in the GAS PR. field. When Dynamic is selected, this setting is not allowed.
- GAS PR. Field: sets the span gas pressure value inside the Audit Cell when the Static mode is selected.

OUTSIDE PATH section:

The optical path length which should be taken into account in the measurement is the length of the Audit Cell. Since the total laser path length is longer because of the flanges, the transmitter, and the receiver, it is necessary to take into account the path length of the flanges, the transmitter and the receiver on both sides of the Audit Cell and remove it from the concentration calculation.

Consequently, the user can set in this section all the necessary information about the optical path due to flanges, transmitter and receiver which will be used in calculations.

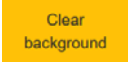
- CONCENTRATION field: sets the concentration value (in % or RH) to be removed from the measurement because of the flange, transmitter and receiver pathlength.
- PATHLENGTH field: sets the optical path length due to the flanges, transmitter and receiver only.
- TEMPERATURE and PRESSURE fields: sets the temperature and pressure values at flange, transmitter and receiver level.

When all the Concentration, Pathlength, Pressure and Temperature fields are set, the user clicks on the button  to obtain the fitted measurement of the desired concentration.

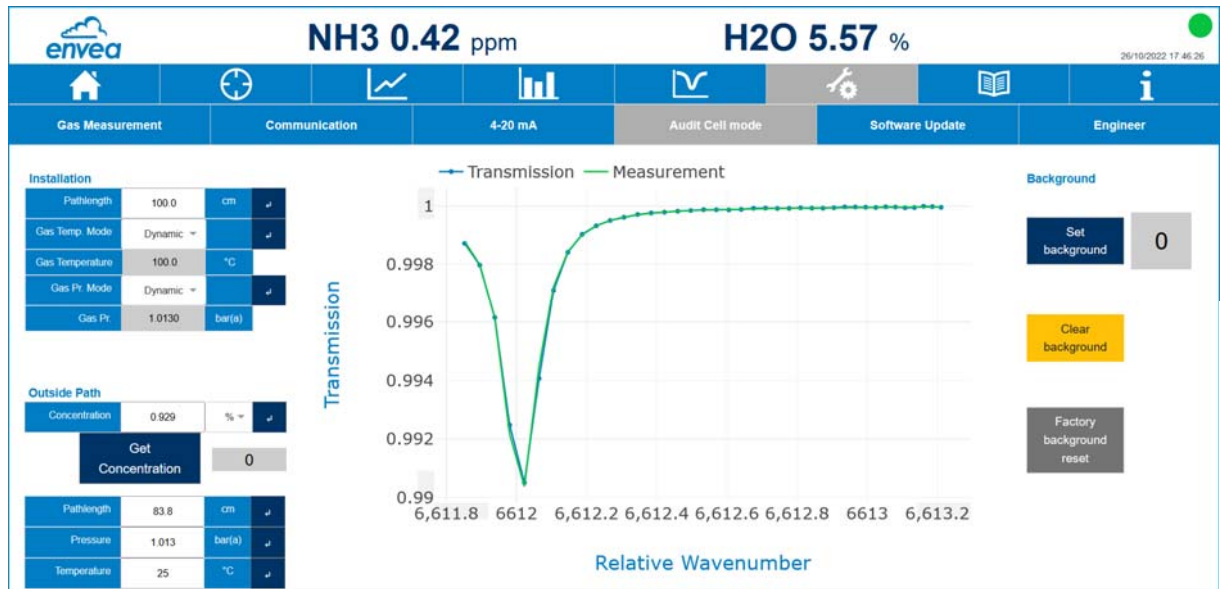
BACKGROUND section :

- SET BACKGROUND button  : the user injects dry air or zero gas, then clicks on this button to set the background. A window opens which asks « Are you sure to set the background? Confirm/Close ». Clicking on « Confirm » sets the background and closes the window.

The analyzer fits the measurement compared to the background it obtained, and the difference between the transmission curve and the measurement curve allows to define the concentration.

- CLEAR BACKGROUND button  it allows to cancel the background values. Consequently, the measurement is not fitted anymore.
- FACTORY BACKGROUND RESET button  : it allows to recover the values set in factory before delivery.

The curve display has the same functionalities as explained in the SPECTRA page.



3.2.6.5. Software update

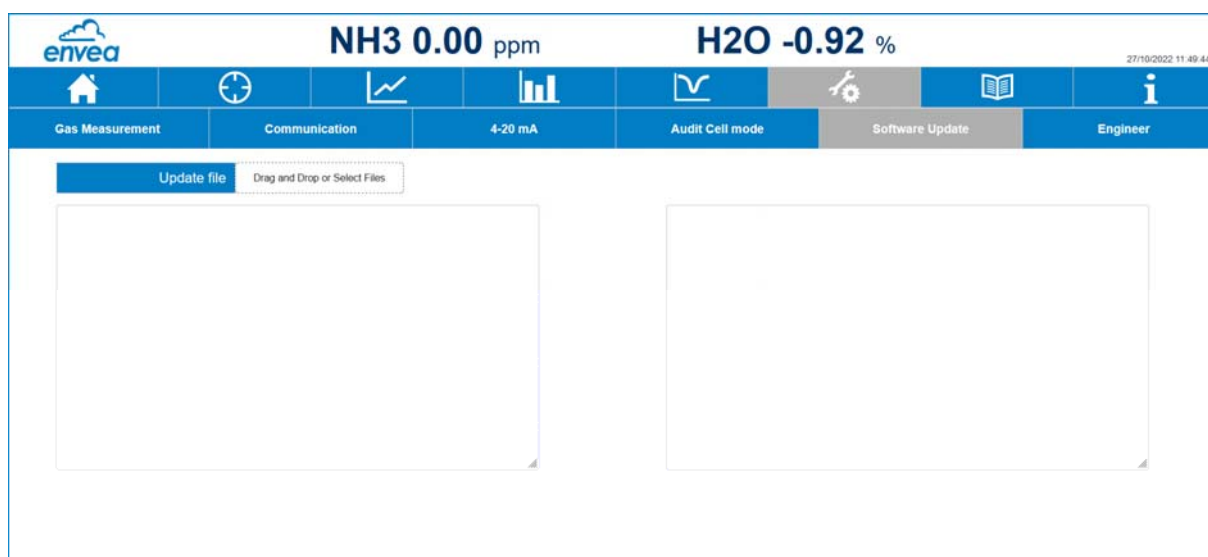
The user displays this sub-menu by clicking on  which becomes . It allows to update this WebServer software. The software update file should be requested from ts@envea.global. It is named "update.sh".

The user drag and drop the "update.sh" file in the UPDATE FILE field, and waits for its loading in the analyzer. When it is finished, two fields are displayed, MASTER PARAMETERS and SLAVE PARAMETERS fields, allowing to update separately the Master and the Slave by filling in the appropriate data and clicking on the UPDATE (navy blue) button.

Otherwise, by default, the user clicks on the UPDATE (navy blue) button to update both the Master and the Slave.

Once update is launched, the analyzer stops while updates are done, which may last some minutes. The user can follow the updating with the specific information displayed in the left window for the Master and the right window for the Slave.

When update is finished, the analyzer will reboot without any action required from the user.



3.2.6.6. Engineer

The user displays this sub-menu by clicking on  which becomes . This access is for ENVEA Service Engineers only.

3.2.7. MANUAL LOADING

The user loads the technical manual of the LAS 5000XD by clicking on  which becomes  when loading is achieved.

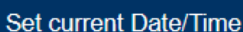
3.2.8. INFORMATION PAGE

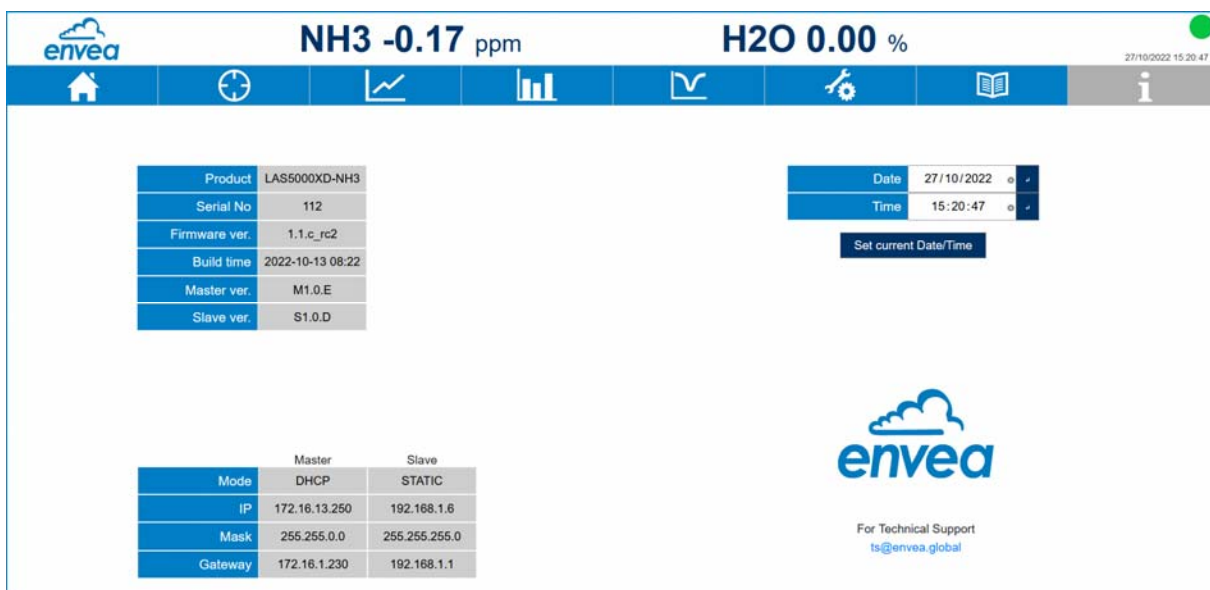
The user displays this page by clicking on  which becomes . It gives the information necessary to identify the analyzer and to communicate with.

The identification information is as follows : the analyzer name with the gas type to be analyzed, its serial number, the firmware version which performs calculations and its build time (i.e. compilation time), the versions of Master/Slave software.

The information necessary to configure the automatic communication of Master and Slave to the network is as follows : the operating mode (DHCP or Static), and their respective IP address, mask and gateway.

NOTE : The Ethernet cable can be connected either on the Master or on the Slave.

The Date and Time fields are used to set the analyzer date and time to the computer date and time on which it is connected. The user clicks on the  button to fill in the Date and Time fields with the current dates of the computer. Then he validates each field with  to set (force) the analyzer to these programmed dates. If not validated, the analyzer returns to its current date and time.



Identification Information:

Product	LAS5000XD-NH3		
Serial No	112		
Firmware ver.	1.1.c_rc2		
Build time	2022-10-13 08:22		
Master ver.	M1.0.E		
Slave ver.	S1.0.D		

Network Configuration:

	Master	Slave
Mode	DHCP	STATIC
IP	172.16.13.250	192.168.1.6
Mask	255.255.0.0	255.255.255.0
Gateway	172.16.1.230	192.168.1.1

Date and Time Settings:

Date	27/10/2022
Time	15:20:47



ENVEA Logo

For Technical Support
ts@envea.global

3.3. CALIBRATION

The LAS 5000XD is set up at the factory to measure and report concentration correctly. To check the accuracy of the LAS 5000XD, there are two available methods: an in-line span check cell or an audit cell.

- In-line span check cell: Install the in-line span check cell on the duct. Test with a flow of a calibrated gas mixture.
- Audit cell: Mount the LAS 5000XD audit cell periodically to check the accuracy of the LAS 5000XD.

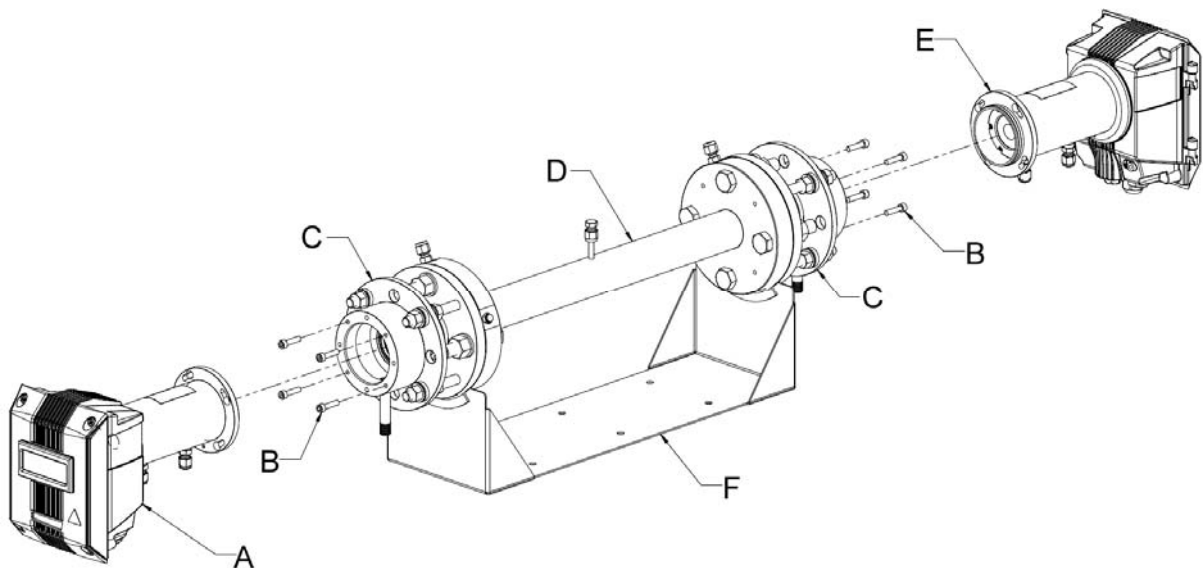
For more information about how often your LAS 5000XD needs calibrating, contact ENVEA or your local ENVEA representative.

3.3.1. AUDIT CELL

The audit cell, in the following image, is made from 316L stainless steel. For reactive gases (e.g. NH_3 or HCl) there is a fluoropolymer coated option. The audit cell can include gas temperature and pressure sensors. The temperature and pressure sensors are optional but will enable the most accurate results.

3.3.1.1. Presentation

The audit cell is presented here-below :



Legends : (A) Transmitter, (B) Set screws, (C) Alignment bellows, (D) Audit cell as such, (E) Receiver, (F) Audit cell mounting plate.

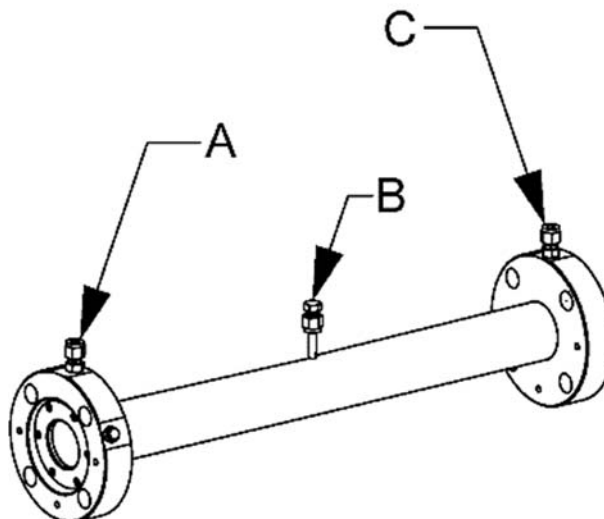
Figure 3–19 – Audit cell presentation with transmitter and receiver

The recommended tools for mounting the audit cell are the followings :

- 6 mm-cross head screwdriver, used to open the lid of the Transmitter and the Receiver.
- 5 mm-Allen key, used for the set screws on the alignment bellows.
- 24 mm-spanner, used for mounting the alignment bellows to the audit cell flange.
- 9/16"-spanner, used for window purge gas connections.

3.3.1.2. Using the audit cell

Gas inlet and outlet connections are fitted on the audit cell flanges (A, C, in the image below). All gas fittings are 1/4" Swagelok. Optional gas fittings, for purging the outside path between the Transmitter / Receiver and cell windows are also located on the LAS 5000XD flanges.



Legends:

- (A) (C) Audit cell flange including gas inlet or outlet. The optional pressure sensor should be mounted on the outlet.
- (B) Optional temperature sensor installation location.

3.3.2. OPTIONAL TEMPERATURE AND PRESSURE SENSORS FOR THE AUDIT CELL

The optional temperature and pressure sensors can be fitted to the audit cell and connected to the LAS 5000XD. For connections, see section 1.1.4.2.

3.3.3. IN-LINE SPAN CHECK CELL

To check the operation and response of the LAS 5000XD an in-line span cell can be used. A known gas concentration is injected into the cell and adds a fixed offset to the reported process concentration. Only non-flammable and non-corrosive gases should be introduced into the in-line span check cell to prevent damage and to prevent a flammable risk.

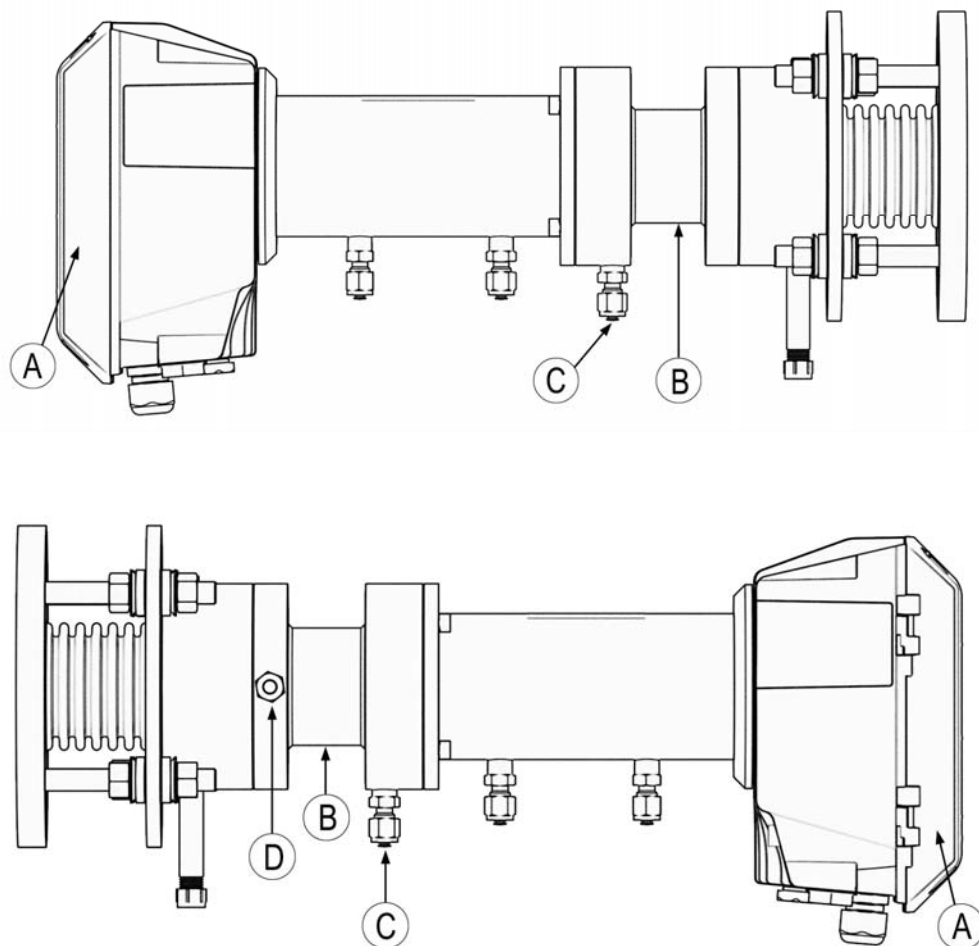
For the span check a cylinder with a concentration of at least 15x the expected process concentration is recommended. This will result in a fixed increase in the reported process concentration. However, the increase will be less than the injected cylinder value.

This difference in concentration is due to the temperature of the span gas as the LAS 5000XD is set to measure at the process temperature and the path length of the span cell is shorter than the process value set in the LAS 5000XD parameters.

As both the process temperature and path length can be set by the user and will depend on the installation, the reported span concentration will vary. For help in estimating the measured span concentration please contact your local ENVEA representative.

3.3.3.1. Presentation

The in-line span cell is presented here-below :



Legend : (A) Receiver, (B) In-line span check cell, (C) Gas in, (D) Gas out.

Figure 3-20 – In-line span cell presentation with receiver

3.3.3.2. Mounting the in-line span cell

Mount the in-line span check cell between the Receiver and the alignment bellows. The following diagram shows an exploded view of in-line span check cell and an LAS 5000XD.

Check the alignment of the LAS 5000XD after installing the in-line cell, see sections 3.1.3.3 and 3.2.2.

The recommended tools for mounting the in-line span cell are the followings :

- 5 mm-Allen key, used for the set screws on the alignment bellows.
- 12 mm-spanner used for gas connections.

3.3.3.3. Using the in-line span check cell

The in-line cell is supplied with $\frac{1}{4}$ " stainless steel compression fittings and these can be exchanged for any suitable fitting with a $\frac{1}{8}$ " BSPT (JIS Rc) thread.

Connect the in-line cell to the flow circuit as shown below. The use of PFA tubing (or similar) is recommended. The flow schematic shown provides a method to flow span gas to the in-line cell and then purge with instrument air. The instrument air should be dry, oil and particulate free.

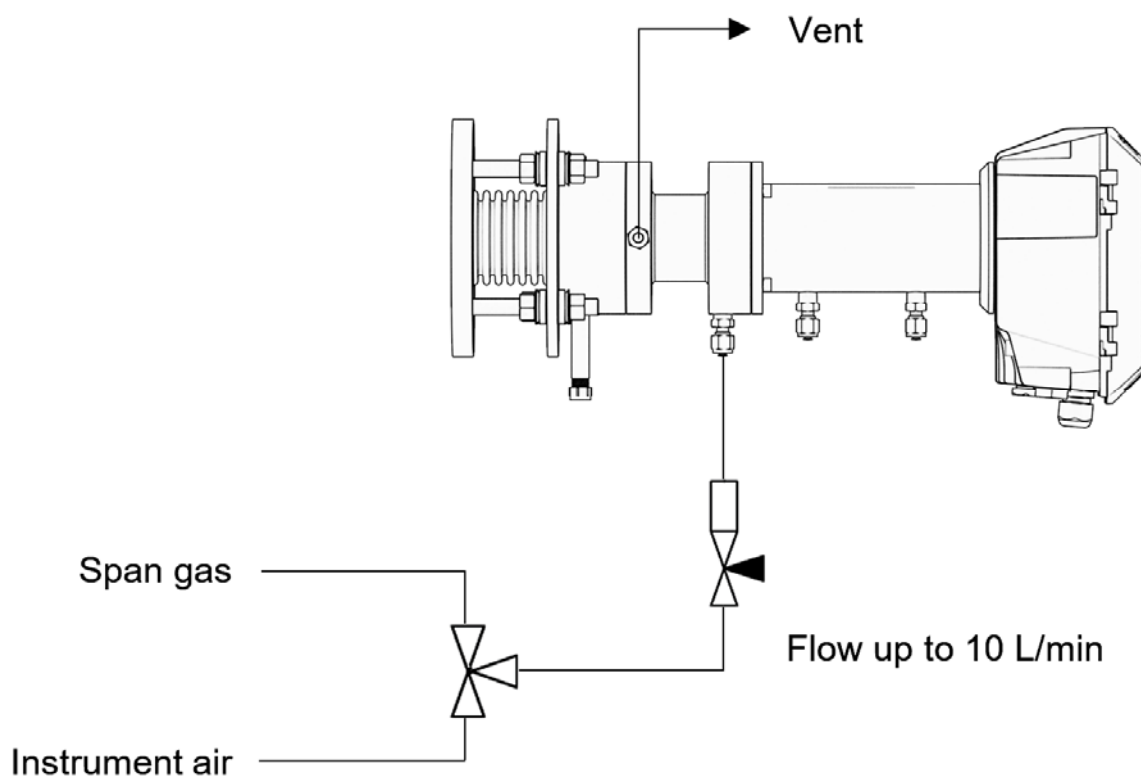


Figure 3–21 – In-line span cell operation

The recommended minimum flow rate should be 4 L/min, the flow rate is dependent on the lengths of tubing required for the installation and the span gas used (NH_3 and HCl may require higher flow rates).

The reported span concentration will vary with the process concentration so the span check should coincide with a stable process period if possible.

The time required to reach a stable level will depend on the tubing length, span gas and the ambient temperature but a minimum of 5 minutes is recommended. The span/purge process can be repeated by turning the 3-way valve as required. The time interval between span checks can vary and for assistance please contact your local ENVEA representative.

The span cell should be purged with instrument air for a minimum of 15 minutes after the final span check. The flow rate can then be reduced to zero.

4. CORRECTIVE MAINTENANCE

The LAS 5000XD provides reliable and low-maintenance operation. Occasionally, it could be necessary to maintain the Transmitter and Receiver. The frequency of each task is specific to each installation. For more information, contact ENVEA or your local ENVEA representative.

4.1. WINDOW CLEANING

Process windows protect the LAS 5000XD. The window is continuously purged during normal operation to prevent the build-up of contaminant. The LAS 5000XD continually measures and reports received optical power and the % transmission. Track either of these values using either the LCD or data logging software to identify when the windows need cleaning.

**WARNING!****INVISIBLE LASER RADIATION!**

The LAS 5000XD uses lasers emitting light in the near-infrared region of the spectrum. Power off the LAS 5000XD before disassembling it in any way.

Proceed in the following order :

- Power off the LAS 5000XD.
- Use a 5 mm Allen key to loosen the four bolts to detach the Transmitter from the alignment bellows.
- Rotate and pull the Transmitter or Receiver away from the alignment bellows.
- Carefully put the Transmitter or Receiver to one side, making sure the exposed window A is kept free from contamination.
- Use an air duster, suitable for optics cleaning, to remove loose dust from the window.
- To remove further dust, use isopropanol, $\geq 95\%$ to clean the window. Rinse with de-mineralized water and blow with clean dried air.
- Repeat steps four and five until the window is clean.

If this method does not remove contaminants, replace the window. Contact ENVEA or your local ENVEA representative.

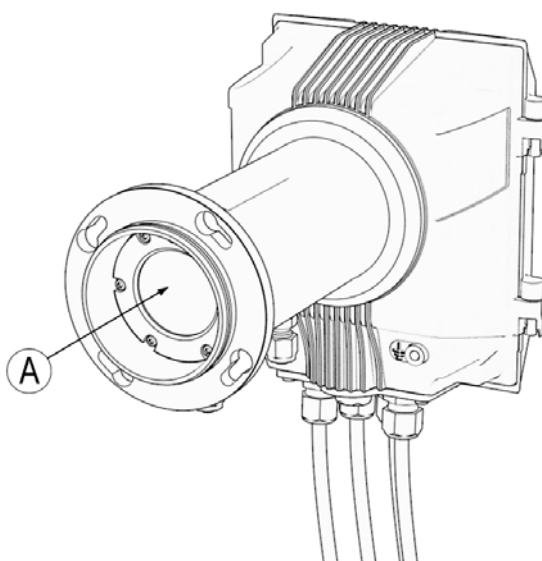


Figure 4–1 – Window cleaning

4.2. OTHER CHECKS



WARNING!
INVISIBLE LASER RADIATION!
The LAS 5000XD uses lasers emitting light in the near-infrared region of the spectrum. Power off the LAS 5000XD before disassembling it in any way.

Step	Details	Reference
Alignment check	Over time, due to external factors, the Transmitter and Receiver’s alignment can change. The %T value on the LCD screen will reduce. Periodically check the Transmitter and Receiver alignment and realign them if necessary.	See sections 3.1.3.3 and 3.2.2 .
Calibration check	Check the calibration of the LAS 5000XD. Two tools are available for checking calibration: the audit cell and the in-line span check cell. Contact ENVEA or your local ENVEA representative for details on how often to check calibration.	See section 3.3.
Clear view check	Over time material can obscure or block the hole through the duct flange. The %T value on the LCD screen will reduce. Periodically remove the Transmitter and Receiver from the alignment bellows and inspect the duct flange hole. Clean if necessary.	

4.3. ANALOG INPUT/OUTPUT CONFIGURATION

4.3.1. ANALOG OUTPUT CONFIGURATION

The analog outputs are assigned to the gas concentration measurement. They are configured to convert the concentration measurements into 4-20 mA signals and allow the data numeric communication. By default, the analog output 1 is configured for the gas n°1, and the analog output n°2 is configured for the gas n°2.

The analog output configuration consists in matching the 4 mA to the minimum value of the gas concentration range, and the 20 mA to the maximum value of the gas concentration range. The procedure to be applied is the same for the analog outputs 1 and 2.

The analog output n° 1 configuration is detailed below as example. To do that, a mA/A-set multimeter is necessary.

Proceed in the following order :

1 – Setting the analog output 1 to 4 mA, i. e. the minimum value of the gas concentration range

In the ANALOG OUTPUT 1 section, set the TEST MODE field to ON and click on the  button to confirm. Enter 4 mA in the TEST VALUE field and click on  to confirm, or press down the  button to directly set the minimum range value to 4 mA. It is therefore necessary to set the analog output 1 of the JUNCTION BOX to match the 4 mA with the MIN DAC VALUE of the corresponding analog output.

To do this, locate this output on the JUNCTION BOX (out ana 1, in this case). Then, using a mA/A-set multimeter (1), place the black test probe on the GND (2) and the red test probe on the output-ana 1 (3). The reading on the multimeter should equal 4 mA (4).

If this is not the case, it is necessary to decrease or increase the value of the MIN DAC VALUE field and click on  to confirm this new value. The measurement with the multimeter has to be renewed until reaching the value of 4 mA on the multimeter (4).



(1) 4-20 mA-configured multimeter, (2) GND, (3) analog output 1, (4) 4 mA reading on multimeter.

Figure 4-2 – Analog output configuration to minimum value of gas concentration range

2 – Setting the analog output 1 to 20 mA, i. e. the maximum value of the gas concentration range

In the ANALOG OUTPUT 1 section, set the TEST MODE field to ON and click on the  button to confirm. Enter 20 mA in the TEST VALUE field and click on  to confirm, or press down the **Set Max.** button to directly set the maximum range value to 20 mA. It is therefore necessary to set the analog output 1 of the JUNCTION BOX to match the 20 mA with the MAX DAC VALUE of the corresponding analog output.

To do this, locate this output on the JUNCTION BOX (out ana 1, in this case). Then, using a mA/A-set multimeter (1), place the black test probe on the GND (2) and the red test probe on the output-ana 1 (3). The reading on the multimeter should equal 20 mA (4).

If this is not the case, it is necessary to decrease or increase the value of the MAX DAC VALUE field and click on  to confirm this new value. The measurement with the multimeter has to be renewed until reaching the value of 20 mA on the multimeter (4).

3 – Carry out a mid-scale check.

Enter 12 mA in the TEST VALUE field and click on  to confirm. Check that the value read on the multimeter corresponds to the mid-scale value.



(1) 4-20 mA-configured multimeter, (2) GND, (3) analog output 1, (4) 100 % reading on multimeter.

Figure 4-3 – Analog output configuration to maximum value of gas concentration range

4.3.2. ANALOG INPUT CONFIGURATION

See Figure 4–4.

The analog inputs 16 and 17 of the JUNCTION BOX are assigned to the temperature measurements, and the analog inputs 18 and 19 are assigned to the pressure measurements (1). They are configured to convert the values obtained from the temperature and pressure sensors into 4-20 mA signals, in order to be taken into account and processed by the system.

The analog inputs configuration consists in matching the 4 mA with the minimum value of the corresponding temperature or pressure ranges, and the 20 mA with their maximum values. The procedure to be applied is the same for the temperature analog inputs and the pressure analog inputs.

The temperature analog inputs configuration is detailed below. To do that, a 4-20 mA current generator is necessary, it should be able to generate 4, 12 and 20 mA.

Proceed in the following order :

1 – Setting the temperature analog inputs 16 and 17 to 4-20 mA, i. e. the minimum and the maximum temperature range values

Connect the (–) polarity wire of the 4-20 mA current generator to the (16) temperature (–) analog input 4-20 mA, and the (+) polarity wire to the (17) temperature (+) analog input 4-20 mA. Switch on the 4-20 mA current generator.

- Set the 4-20 mA current generator to 4 mA. In the TEMPERATURE section of the CONFIGURATION/ GAS MESUREMENT screen, set the GAS TEMP. MODE field to DYNAMIC by clicking on . Then, go to the 4-20 mA screen : in the ANALOG INPUT 1 : TEMPERATURE section, wait for the TEST VALUE field to decrease and stabilize to zero.

If it is not the case, it is necessary to decrease or increase the value of the MIN ADC VALUE field until reaching zero. Then click on  to confirm.

- Set the 4-20 mA current generator to 20 mA. Wait for the TEST VALUE field to increase and stabilize to the maximum of the full scale, i.e. 200°C for the temperature. If it is not the case, it is necessary to decrease or increase the value of the MAX ADC VALUE field until reaching the maximum of the full scale, i.e. 200°C here. Then click on  to confirm.
- Carry out a mid-scale check. Set the 4-20 mA current generator to 12 mA, and check that the value read in the TEST VALUE field correspond to the mid-scale value.

2 – Setting the pressure analog inputs 18 and 19 to 4-20 mA, i. e. the minimum and the maximum pressure range values

Connect the (–) polarity wire of the 4-20 mA current generator to the (18) pressure (–) analog input 4-20 mA, and the (+) polarity wire to the (19) pressure (+) analog input 4-20 mA. Switch on the 4-20 mA current generator.

- Set the 4-20 mA current generator to 4 mA. In the PRESSURE section of the CONFIGURATION/ GAS MESUREMENT screen, set the GAS PR. MODE field to DYNAMIC by clicking on . Then, go to the 4-20 mA screen : in the ANALOG INPUT 2 : PRESSURE section, wait for the TEST VALUE field to decrease and stabilize to zero.

If it is not the case, it is necessary to decrease or increase the value of the MIN ADC VALUE field until reaching zero. Then click on  to confirm.

- Set the 4-20 mA current generator to 20 mA. Wait for the TEST VALUE field to increase and stabilize to the maximum of the full scale, i.e. 2.5000 bar for the pressure. If it is not the case, it is necessary

to decrease or increase the value of the MAX ADC VALUE field until reaching the maximum of the full scale, i.e. 2.5000 bar here. Then click on  to confirm.

- Carry out a mid-scale check. Set the 4-20 mA current generator to 12 mA, and check that the value read in the TEST VALUE field correspond to the mid-scale value.



Figure 4-4 – Analog input configuration

4.4. MAINTENANCE KITS AND RECOMMENDED SPARE PARTS

The maintenance kits are as follows:

Designation	References
Maintenance kit for ACCAIR-B-Y	SPABLW-S-FLT-V2
Maintenance kit for MDS	MDS-K1
Maintenance kit for F05-0394-A	F05-K-0394-A

The recommended spare parts are as follow:

– RSP kit for LAS 5000XD (Ref. LAS5000XD-RSP):

Line nb	Designation	Reference	Qty
1	Mesh filter	F05-4002	2
2	O-ring 44x2,5	G06-044_0-2_5-S	2
3	O-ring 53,57x3,53	G06-053_57-3_53-S	2
4	O-ring 71x2,5	G06-071_0-2_5-S	4
5	M3 socket screw - Length 6mm	G07-CC-03-006-I	12
6	M6 socket screw - Length 25mm	G07-CC-06-025-I	8

– RSP Window kit for LAS 5000XD Standard (Ref. LAS5000XD-RSP1):

Line nb	Designation	Reference	Qty
1	O-ring 44x2,5	G06-044_0-2_5-S	2
2	M3 socket screw - Length 6mm	G07-CC-03-006-I	12
3	Window Assembly	P10-4175-A-SAV	2

– RSP Window for LAS 5000XD HF (Ref. LAS5000XD-RSP2):

Line nb	Designation	Reference	Qty
1	O-ring 44x2,5	G06-044_0-2_5-S	2
2	M3 socket screw - Length 6mm	G07-CC-03-006-I	12
3	HF Window Assembly	P10-4180-A-SAV	2

5. ABBREVIATIONS

Accessory	Note
%	Percentage
°C	Degrees Celsius
AI	Analog input
AO	Analog output
ASME	American Society of Mechanical Engineers
Com	Communications
BSP	British standard pipe
Hz	Hertz
I_t	Received optical intensity
I_o	Transmitted optical intensity
kg	Kilogram
l	Path length
LCD	Liquid crystal display
LED	Light emitting diode
mA	Milliamperes
mm	Millimetre
Modbus RTU	Modbus remote terminal unit
PCB	Printed circuit board
PCD	Pitch circle diameter
ppm	Parts per million
RX	Receiver
TX	Transmitter
USB	Universal serial bus
V DC	Volts of direct current
W	Watts

6. APPENDIX

6.1. MODBUS MAPPING

The official Modbus specification can be found at www.modbus.org/specs.php.

6.1.1. GLOSSARY

Parameter	Description
Address	Integer address of the location in the memory map (word format => 2 bytes)
R	Read-only parameter
R/W	Read/write parameter
W	Write-only parameter
String	20-byte character string using ten Modbus registers
Float	IEEE754 encoded floating point number using two Modbus registers
Register	Register of 16 bits
Word	16-bit unsigned integer using one Modbus register
Unsigned Int	32-bit using two Modbus registers

6.1.2. FUNCTIONS CODES

Code	Action
03	Read holding registers
06	Write single holding register
16	Write multiple holding registers

6.1.3. MODBUS MAPPING

- Parameters using multiple registers must be read, or written to, with a single command. The LAS 5000XD reports an error if registers are read, or written to, with multiple commands.
- When interpreting values, please note that values are Big-Endian when transmitted.
- The Modbus protocol used is as specified in *Modbus Application Protocol Specification V1.1B3, P7, section 4.4.*
- If you try to write to a read-only address, no change takes place and no error is reported.

NOTE : Parameters **must** be written within the Min and Max boundaries, as shown in the last two columns of the table below.

Table 6-1 – Modbus mapping

Start Address	End	Access	Data	Size Number of modbus registers	Parameter	Description	Example	Minimum value allowed	Maximum value allowed
1000	1009	R	String	10	Instrument ID	Instrument ID	LAS5000XD-NH3		
1010	1019	R	String	10	Serial Number	Serial number	40		
1020	1029	R	String	10	Firmware version	Firmware version	1.0.a		
1030	1039	R	String	10	Slave firmware version	Slave firmware version	10,23		
1040	1041	R	Float	2	Internal temperature	Internal temperature	25		
1042	1043	R	Float	2	Slave internal temperature	Slave internal temperature	25		
2000	2001	R	Float	2	Gas temperature in °C	Gas temperature in °C	50		
2002	2003	RW	Float	2	Gas temperature setting	Gas temperature setting	50		
2004	2004	RW	Word	1	Gas temperature type	Gas temperature type (static / dynamic)	0		
2005	2006	RW	Float	2	Gas temperature process min setting	Gas temperature process min setting	0		
2007	2008	RW	Float	2	Gas temperature process max setting	Gas temperature process max setting	0		
2009	2010	RW	Float	2	Gas temperature min setting	Gas temperature min setting	0		
2011	2012	RW	Float	2	Gas temperature max setting	Gas temperature max setting	50		

Start Address	End	Access	Data	Size Number of modbus registers	Parameter	Description	Example	Minimum value allowed	Maximum value allowed
2013	2014	RW	Float	2	Gas temperature min ADC value	Gas temperature min ADC value	75		
2015	2016	RW	Float	2	Gas temperature max ADC value	Gas temperature max ADC value	3500		
2017	2018	RW	Unsigned int	2	Gas temperature average count	Gas temperature average count			
2100	2101	R	Float	2	Measure pressure	Measure pressure	0,999	??	??
2102	2103	RW	Float	2	Gas pressure setting	Gas pressure setting	0,0001	??	??
2104	2104	RW	Word	1	Gas pressure type	Gas pressure type (static / dynamic)	0	??	??
2105	2106	RW	Float	2	Gas pressure process min setting	Gas pressure process min setting	0,001	??	??
2107	2108	RW	Float	2	Gas pressure process max setting	Gas pressure min process setting	10	1	??
2109	2110	RW	Float	2	Gas pressure min setting	Gas pressure min setting	0	1	??
2111	2112	RW	Float	2	Gas pressure max setting	Gas pressure max setting.	2,5		
2113	2114	RW	Float	2	Gas pressure min ADC setting	Gas pressure min ADC setting	45		
2115	2116	RW	Float	2	Gas pressure max ADC setting	Gas pressure max ADC setting	3990		
2117	2118	RW	Unsigned int	2	Gas pressure average count	Gas pressure average count			
3000	3009	R	String	10	Gas n°1 name	Gas n°1 name	NH3		
3010	3019	R	String	10	Gas n°1 unit	Gas n°1 unit	ppm		
3020	3021	R	Float	2	Gas n°1 concentration	Gas n°1 concentration			
3022	3023	R	Float	2	Gas n°1 calibrated concentration	Gas n°1 calibrated concentration			
3024	3025	R	Float	2	Gas n°1 raw concentration	Gas n°1 raw concentration	0		
3026	3027	RW	Float	2	Gas n°1 concentration coeff.	Gas n°1 concentration coeff.	0		

Start Address	End	Access	Data	Size Number of modbus registers	Parameter	Description	Example	Minimum value allowed	Maximum value allowed
3028	3029	RW	Float	2	Gas n°1 concentration offset	Gas n°1 concentration offset	4		
3030	3031	RW	Unsigned int	2	Gas n°1 rolling average	Gas n°1 rolling average	11	1	??
3100	3109	R	String	10	Gas n°2 name	Gas n°2 name	H2O	??	??
3110	3119	R	String	10	Gas n°2 unit	Gas n°2 unit	%	??	??
3120	3121	R	Float	2	Gas n°2 concentration	Gas n°2 concentration		??	??
3122	3123	R	Float	2	Gas n°2 calibrated concentration	Gas n°2 calibrated concentration		??	??
3124	3125	R	Float	2	Gas n°2 raw concentration	Gas n°2 raw concentration		??	??
3126	3127	RW	Float	2	Gas n°2 concentration coeff.	Gas n°2 concentration coeff.	1	0	??
3128	3129	RW	Float	2	Gas n°2 concentration offset	Gas n°2 concentration offset	1	0	??
3130	3131	RW	Float	2	Gas concentration rolling average	Gas concentration rolling average	11	1	??
4000	4000	RW	Word	1	Analog output n°1 unit	Analog output n°1 unit	2	0	5
4001	4002	RW	Float	2	Analog output n°1 test value	Analog output n°1 test value	0	0	1
4003	4004	RW	Float	2	Analog output n°1 min value	Analog output n°1 min value	0	??	??
4005	4006	RW	Float	2	Analog output n°1 max value	Analog output n°1 max value	50	??	??
4007	4008	RW	Float	2	Analog output n°1 min DAC value	Analog output n°1 min DAC value	53	0	4000
4009	4010	RW	Float	2	Analog output n°1 max DAC value	Analog output n°1 max DAC value	3880	0	4000
4100	4100	RW	Word	1	Analog output n°2 unit	Analog output n°2 unit	2	0	5
4101	4102	RW	Float	2	Analog output n°2 test value	Analog output n°2 test value	5,32	??	??
4103	4104	RW	Float	2	Analog output n°2 min value	Analog output n°2 min value	1	??	??
4105	4106	RW	Float	2	Analog output n°2 max value	Analog output n°2 max value	30	??	??

Start Address	End	Access	Data	Size Number of modbus registers	Parameter	Description	Example	Minimum value allowed	Maximum value allowed
4107	4108	RW	Float	2	Analog output n°2 min DAC value	Analog output n°2 min DAC value	60	0	4000
4109	4110	RW	Float	2	Analog output n°2 max DAC value	Analog output n°2 max DAC value	3800	0	4000
5000	5001	R	Float	2	Laser temperature	Laser temperature	3	??	??
5002	5003	R	Float	2	Global gain	Global gain	1	??	??
5004	5005	RW	Float	2	Fixed global gain	Fixed global gain	1	1	128
5006	5006	RW	Word	1	Auto gain	Auto gain	1	0	1
5007	5008	R	Float	2	Percentage light power	percentage light power	1	??	??
5009	5010	R	Float	2	Zero power	Zero power	2	??	??
5011	5012	R	Float	2	Normalized light power	Normalized light power		??	??
5013	5014	RW	Float	2	Photoreceiver thresholds reference value	Photoreceiver thresholds reference value	8		
5015	5016	R	Float	2	Raw light power	Raw light power		??	??
5017	5018	R	Float	2	Standard deviation first fit	standard deviation first fit		??	??
5019	5020	R	Float	2	Niter first fit	Niter first fit		??	??
5021	5022	R	Float	2	Laser x-axis tolerance	Laser x-axis tolerance		??	??
5023	5024	RW	Float	2	Instrument pathlength	Instrument pathlength	23		
5025	5026	RW	Float	2	Outside path pathlength	Outside path pathlength	23		
5027	5028	RW	Float	2	Outside path VMR	Outside path VMR	23	0	1
5029	5030	RW	Float	2	Outside path pressure	Outside path pressure	50	0	1
5031	5032	RW	Float	2	Outside path temperature	Outside path temperature	1,2		
5033	5034	RW	Float	2	Align	Align		??	??

Start Address	End	Access	Data	Size Number of modbus registers	Parameter	Description	Example	Minimum value allowed	Maximum value allowed
5100	5109	RW	String	10	Date	Date			
5110	5119	RW	String	10	Time	Time			

6.2. ERROR CODES

6.2.1. LAS 5000XD DISPLAY ERROR CODES

The following errors appear on the display of the LAS 5000XD:

Error nb	Message displayed
(1)	"Error : Slave unit not responding (1)"
(2)	"Error : UDP listener (2)"
(4)	"Error : MODBUS (4)"
(8)	"Info : Analyzer reconfiguring, please wait..."
(16)	"Error : Low laser power (16)"
(32)	"Error : Laser temperature (32)"
(64)	"Error : Gas pressure sensors (64)"
(128)	"Error : Gas temperature sensor (128)"
(256)	"Error : Laser ramp saturation (256)"
(512)	"Error : Configuration error (512)"
(1024)	"Error : ADC saturation (1024)"
(2048)	"Error : Long UDP listener (2048)"
(4096)	"Error : PGA saturation (4096)"
(8192)	"Error : Internal temperature (8192)"
(16384)	"Error : Fit parameter at limit (16384)"

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